





Final Year Project Report

Centralized security configuration scanning tool to detect ransomware-related misconfigurations in SME environments

By

TEH JUN HAO

TP067556

APU3F2511CS(CYB)

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at Asia Pacific University of Technology and Innovation.

Supervised by: DR. Hassan Jamil Syed

2nd Marker: Yogeswaran A/L Nathan

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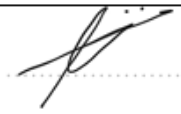
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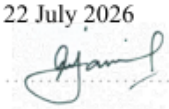
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Title of the Final Year Project / Dissertation / Thesis : **Centralized security configuration scanning tool to detect ransomware-related misconfigurations in SME environments**
Abstract : Ransomware attacks have become one of the most significant cybersecurity threats affecting Small and Medium Enterprises (SMEs). Many SMEs lack the resources and expertise to continuously monitor their security configurations, making them vulnerable to ransomware-related misconfigurations. This project presents a centralized security configuration scanning tool that remotely scans both Linux and Windows systems to detect security misconfigurations associated with ransomware attacks. The system calculates a risk score based on the identified findings and provides remediation recommendations through a centralized web dashboard and downloadable PDF reports. The developed solution enables administrators to proactively identify security weaknesses, improve security posture, and reduce the risk of ransomware incidents in SME environments.
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Abstract

Ransomware stands as one of the most severe cybersecurity risks facing companies in the world. Various ransomware attacks take place not just due to the existence of skilful malware but also because of improper system settings, such as remote services exposed to the Internet, weak password policies, unpatched systems, and unauthorized access. Small and medium enterprises are the most endangered on this front since they do not have optimal cybersecurity teams and centralized security management systems.

This paper presents the idea of developing and testing a centralized security configuration scanner that allows for the identification of system settings that are misconfigured concerning ransomware threats. The platform is meant to scan, on separate machines, multiple devices in a network using secure connection technologies: WinRM or SSH. The program determines the configuration, assigns risk groupings such as High, Medium, or Low, and provides remediation recommendations in the reports.

A cross-sectional data collection method is employed to establish how SME IT administrators set up their systems for current security procedures. The results will be used to develop the system requirements and tool development process. The foreseeable result of this project is that SMEs will have a lightweight, user-friendly tool that detects spikes in ransomware threats among their configurations and addresses them accordingly.

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Chapter 1

1.1 Introduction

Modern businesses cannot operate without information systems, as these systems allow operations, communication, and data storage. With increasing complexity, secure system configurations are part of the security methodology of system management. Secure configuration entails password policies, access levels, firewall settings, software updates, and logging.

If system configurations are left unmanaged, organizations risk security breaches. Ransomware is one of the greatest cybersecurity threats affecting organizations today (ENISA, 2023; Verizon, 2023), causing significant financial and operational damage worldwide (ENISA, 2023). Many ransomware attacks happen because systems are not configured properly: remote access services exposed and unpatched systems (Kharraz et al., n.d.; Sgandurra et al., 2016).

Small and medium-sized enterprises (SMEs) rarely have dedicated security teams and centralized security systems, which makes them vulnerable to such attacks. For this reason, this project would like to build and test a centralized security configuration scanning tool for detecting ransomware-related misconfigurations in SME setups.

1.2 Problem Background

Many small and medium enterprises (SMEs) find it challenging to ensure safe configuration of their systems. Limited resources, such as budget, workforce, and IT knowledge, make it impossible for many SMEs to employ centralized security management solutions for the task. Instead of relying on tools that automate the task, they conduct manual or ad-hoc checks on security. Over time, there is a risk of configuration drift and concealed threats, when system configurations are altered and left insecure.

Please also refer to existing case studies to understand that ransomware attacks are rising, occurring predominantly in SMEs. Many ransomware occurrences take place because of system misconfiguration, such as RDP exposed to a network, weak passwords, unpatched

systems, administrative rights, and firewall synchronization set up incorrectly. Due to these vulnerabilities, hackers get entry to the network and circulate before the ransomware attack.

Security scanning tools are available on the market; however, they are typically enterprise-based. These tools are large suites that can be time-consuming and costly for SMEs to deploy and manage. Therefore, to respond to the SMEs' configuration scan needs, there is a requirement for a simplistic solution that focuses on ransomware and one that can be easily integrated in the SME environment.

1.3 Project Aim

To create, implement, and test a scanning tool that identifies potential misconfigurations for ransomware that are relevant in a networked environment and refines practical remediation instructions along with proving that risk is reduced by test runs that vary the practice.

1.4 Objectives

- To develop a modular architecture that covers specification input, remote connection with authentication components, individual configuration checks, scoring elements, and report production capabilities.
- To carry out centralized remote scanning from the scanning machine to several target systems using WinRM and PowerShell Remoting for Windows machines and/or SSH for Linux systems with the aim of gathering read-only evidence.
- To refine a risk scoring method that produces High, Medium, and Low priorities for serendipity as well as PASS, FAIL, or INFO outcomes for each security check.
- To establish a controlled network environment that is virtualized and characterize various security misconfigurations for post-scanning manipulation of the tool.

1.5 Scope

The project is about presenting a design and implementation of a centralized security configuration scanning tool in a virtual lab using a controlled environment.

The following aspects of the project are covered:

- Ransomware common misconfiguration types (e.g., remote access exposure, password policy, patch status, permit count, firewall settings, logging configuration).
- Creating a centralized scanning system using remote connection options (e.g., WinRM or SSH).
- Implementing High, Medium, and Low-level risk classifications.
- Creating simple reports with recommendations for remediation.
- The system is assessed with before-and-after tests in the virtual lab.

The project omits certain elements:

- Actual rollout into a live location of a real organization.
- Building malware or ransomware in real running production systems.
- Full-fledged development of an enterprise vulnerability scanner.

1.6 Potential Benefit

The centralized security configuration scanning tool can be used to gain a range of benefits, especially in small and medium-sized enterprises.

1.6.1 Tangible Benefits

To begin with, it enables users to identify potential system abuses leading to ransomware attacks. The security administrator does not have to go through a long manual process in detecting the security threats because the system automatically scans configurations for items such as password policies' patch status, exposure's remote access, firewall settings, and logging configurations. This allows administrators to immediately identify the lead security threat.

The tool also offers simple risk scoring coupled with remediation action or suggestions. This makes it easier for the system administrators to prioritize security threats according to severity levels, for instance, High, Medium, or Low. As such, these organizations end up reacting more quickly and bringing down the risk of security incidences.

Aside from this, the tool provides centralized addressing portfolio that enables admins to keep watch over numerous systems from a single scanning machine. Not only does this

eliminate the need for manual inspection, but it also leads to an increase in operational efficiency.

1.6.2 Intangible Benefits

On top of the technical benefits, this project ensures the increase of security awareness among small and medium-sized business owners. Specifically, it helps organizations realize how specific configuration mistakes that can help ransomware attacks, which in turn would help them improve their security practices.

Moreover, this project contributes to research in academic as it shows SMEs environment software configuration scanning to be effective in minimizing this risk.

1.6.3 Target Users

The system is primarily intended for the following groups:

- IT administrators of SMEs
- Administrators who manage security components for an enterprise environment
- Cybersecurity students and researchers who are investigating system configuration security practices

1.7 Overview of the IR

The Investigation Report (IR) is structured as follows, consisting of four chapters.

Chapter 1: Introduction

Introduces the background of the project, problem statement, project aim, objectives, scope, and potential benefits. It elaborates on the importance of secure management of configuration and the motivation for designing a scanning tool for ransomware attacks that specifically target this area.

Chapter 2: Literature Review

This chapter discusses the literature that is directly concerned with the three issues at hand, namely ransomware threats, management of security configuration, and misconfiguration vulnerabilities. Moreover, the current review analyses similar solutions and instruments that are currently available in the market and attempts to identify their advantages and shortcomings as well as gaps left by previous studies.

Chapter 3 Methodology

This chapter covers the process of how the system development methodology was applied by the project. It breaks down the Agile development strategy, the creation of the survey-based data gathering design, and the techniques used to analyse the collected data for deriving the requirements.

Chapter 4: Conclusion

This chapter is concerned with the project investigation phase, and it summarizes the findings and main results of this stage of the inquiry process. It also illustrates the changes that might be considered and the need for further research within the development phase.

1.8 Project Plan

The Investigation Report (IR) project phase plan specifies a sequence of actions and a timeframe to ensure the effective completion of this project phase. Different project stages are conducted so that the research and preparation can be held in an organized way.

The first stage, topic selection and proposal development, refers to the scene from the movie, where the principal character identifies the problem that needs to be solved and gets approval from experts. In this stage, background search is accomplished to consider the nature of ransomware threat and security configuration management methods.

In the second stage, literature is reviewed through the analysis of theoretical and practical security tools and reports. This stage is essential for laying the theoretical groundwork of the paper and recognizing the research gap.

The third stage requires methodology design, which involves specifying the techniques and data collection process. Survey questionnaire is developed to determine information from participants regarding the configuration of security settings and ignorance of ransomware.

After that, data is gathered through surveys and analysed to determine the connection between security issues and user needs in the fourth stage.

At last, Investigation Report is prepared and consolidated in the IR guidelines in the documentation and report preparation stage.

The detailed timeline of the activities is highlighted in the Gantt Chart in Appendix D.

Chapter 2

2.1 Domain Research

2.1.1 Ransomware Threat Landscape

The ransomware ecosystem is today a serious concern to most organizations. When a target is affected with ransomware, its encrypted files or systems are held for ransom to be decrypted. Modern ransomware attacks often involve multiple stages including initial access, privilege escalation, lateral movement, and data encryption (Kharraz et al., n.d. ; Sgandurra et al., 2016)

Studies have revealed that many ransomware attacks have implications beyond just the sophistication of the malware itself, which results from poor security configurations. Vulnerabilities such as exposed Remote Desktop Protocol (RDP), weak passwords, unpatched systems, misleading privileges, and bad firewall configurations are all among the most common access points. It is these vulnerabilities that allow attackers to enter and move around the network easily.Recent industry reports also indicate that ransomware remains one of the most common forms of cybercrime affecting organizations worldwide (Verizon, 2023; Symantec, 2022).

Small and medium enterprises (SMEs) are particularly at risk because they generally don't have the resources to set up dedicated teams or monitoring. Hence, the adoption of security measures as a preventive tool is indispensable to mitigate the ransomware risks.

2.1.2 Security Configuration Management

Security configuration management is the activity of determining and maintaining the system settings in line with security best practices. This involves password management, access control, network services, patching, antivirus, logging and ensures that systems follow secure baseline settings and reduces the likelihood of vulnerabilities (Holm et al., 2011; NIST, 2020).

Security and privacy challenges in modern networked systems have become increasingly complex due to the growth of interconnected devices and services (*View of Security and Privacy Challenges in the Internet of Things*, 2026). Many companies rely on security

baselines like the CIS Benchmarks to ensure that the systems are configured to support the recommended security policies. Regular assessment looks for configuration drift, which happens when the system settings are altered over some time and cause a security vulnerability.

When configuration assessment is consolidated, it allows the system administrator to scan multiple systems remotely and report misconfigurations from a single management interface. Then it becomes clear that centralized scanning is an order of magnitude better from the standpoint of efficiency, uniformity, and oversight over the network than is manual checking.

Despite the fact, many SMEs use manual inspection or irregular reviews because of budget shortage and lack of technical skills.

2.1.3 Relationship Between Misconfiguration and Ransomware

Studies also suggest that misconfiguration of systems is an important aspect of ransomware attacks. For instance:

- Public-facing RDP services render one vulnerable to brute-force attacks.
- Relaxed password requirements allow one's credentials to be compromised.
- Out-of-date systems introduce known threats.
- Far-reaching administrative privileges facilitate lateral movement.
- Deactivated logging compromises its ability to detect malignancy.

This proves that bettering configuration management can lower the chances of success for a ransomware attack. Accordingly, a security configuration scanner that zeroes in on weaknesses regarding ransomware might be of practical relevance to SMEs. This shows that improving system configuration management can significantly reduce the risk of ransomware attacks (Kolodenker, n.d.; Cabaj & Mazurczyk, 2016).

2.2 Similar Systems / Works

2.2.1 Existing Configuration Scanning Tools

Vulnerability scanning tools are commonly used to identify security weaknesses in systems and networks (Holm et al., 2011; Scarfone et al., 2008). There are several tools that can be used to evaluate system security configurations, such as:

- Nessus
- OpenVAS
- CIS-CAT
- Microsoft Security Compliance Toolkit

The tools can do a check for vulnerability scans and verify compliance with baselines. They also provide comprehensive reports and indicate what may be security risk factors. Nevertheless, most of such tools are enterprise-oriented solutions. They can demand advanced installation, a subscription fee, and high technical skills to manage. The solutions can be pricy or even hard to implement in SMEs.

2.2.2 Strengths and Weaknesses of Existing Systems

Strengths:

- Addresses a wide spectrum of vulnerabilities
- Cross-platform support
- Automated reporting
- Adheres to industry-standard benchmarks

Weaknesses:

- Premium functionality is expensive
- Challenging setup
- Not targeted specifically at ransomware-related misconfigurations
- May produce too many technical outputs that are hard for SMEs' admins to read

2.2.3 Research Gap

Despite the availability of many security tools, there are not many that have been developed with the aim of identifying configuration issues that could be directly exploited for ransomware attack pathways. Moreover, there is a gap in addressing the needs for lightweight and pragmatic tools that have been developed specifically for SME use cases.

Therefore, this means that there is a void for creating a unified platform that would serve as a ransomware-focused configuration scanner that is easy to implement, cheap, and will focus on the more common misconfigurations relevant to SMEs.

2.3 Technical Research

2.3.1 System Requirements Analysis

Security testing and assessment techniques are commonly used to evaluate system vulnerabilities and configuration weaknesses (Scarfone et al., 2008). According to gathered data from related material and domain research, the following should be part of the recommended system:

- Centralized scanning facility
- Remote connection tool (WinRM / SSH)
- Updateable modular configurations
- Risk ranking level (High, Medium, Low)
- Reporting support with solutions clarification

The system working environment should be in a fully controlled virtual lab for safe testing.

2.3.2 Hardware and Software Requirements

Hardware Requirements:

- One scanning machine (host system)
- Many virtual machines (target systems)
- Virtual environment with 8GB minimum RAM

Software Requirements:

- Windows / Linux OS
- Scripting with PowerShell or Python
- Virtualizing software (VirtualBox / VMware etc.)
- Report reading through a web browser

Such tools are selected to be inexpensive and commonly used to work in SME environments.

Chapter 3

3.1 System Development Methodology

The project chooses the Agile system development methodology. Agile is selected since it focuses on small iterations to continuously improve, can be guided through changes, and it is not a requirement to define all the precise elements during the design phase.

The development process consists of several phases:

- Requirement Analysis
  Define what the users need from existing literature and a survey.
- System Design
  Design the architecture of the system and specify the modules to check the configuration.
- Implementation
  Develop different scanning modules utilizing scripting tools.
- Testing
  Conduct the tests in a virtual lab environment.
- Evaluation
  Compare the results after remediation with those obtained before.

Agile is suitable for the project since the new configuration may be developed according to the actual test results and feedback.

3.2 Data Gathering Design

The study used a survey questionnaire to collect data relevant to security configuration management practices and ransomware-related awareness among IT-related stakeholders. Survey methods are widely used in cybersecurity research to collect information regarding security practices and user awareness (Bada et al., 2019).

A structured questionnaire was developed and shared online through Google Forms. In total, 38 respondents were involved through the survey. Among the participants were IT administrators, system engineers, cybersecurity personnel, IT support, and students related to IT.

The survey was anonymous, and participations were voluntary. No personally identifiable data were asked to maintain privacy and confidentiality of the respondents. Thus, ethical approval was provided prior to data collection.

The questionnaire was given three main sections:

Section A: Demographic Information

The section asked the participants' gender, age group, their current position, and their experience in IT.

Section B: Security Configuration Practices

The section asked the participants about their practice with security configuration management, including how often they check configurations and whether there are monitoring tools integrated.

Section C: Perception and System Requirements

The section asked the participants related to their awareness about ransomware and their perceived helpfulness and functionalities of a centralized searching tool that scans for configurations.

The information was exported from Google Forms for further processing.

3.3 Analysis

The descriptive statistical methods were used to analyse the survey data. Frequency counts and percentages were used as a summary measure of responses.

The first element of the analysis was to examine the respondents' demographics in terms of gender, age group, occupation, and years of experience in the field of IT. The survey had 38 qualified respondents. Pie charts were used to summarize the demographics to better represent the respondent profile.

The next step was to look at the respondents' answers regarding the configuration practices related to security to understand shared practices. The analysis considered the percentage of responses related to how often system configurations are reviewed, whether a centralized monitoring tool is applied, and whether the respondent had experienced incidents of ransomware attacks.

Likert-scale survey (Strongly Agree to Strongly Disagree) was analysed for each category of response by calculating the percentage of each. The analysis will help understand how respondents perceive a weak system configuration in terms of an increased risk of ransomware attacks and how helpful they consider a centralized configuration scanning tool.

The survey results were revealed through charts and graphs based on the answers answered in Google Forms to show the share of options chosen.

The survey results have led to a set of user requirements. The requirements will help define the design and construction of the suggested centralized ransomware-focused configuration scanning tool in the next step of the project.

Section A: Demographic Information

3.3.1 Gender Distribution

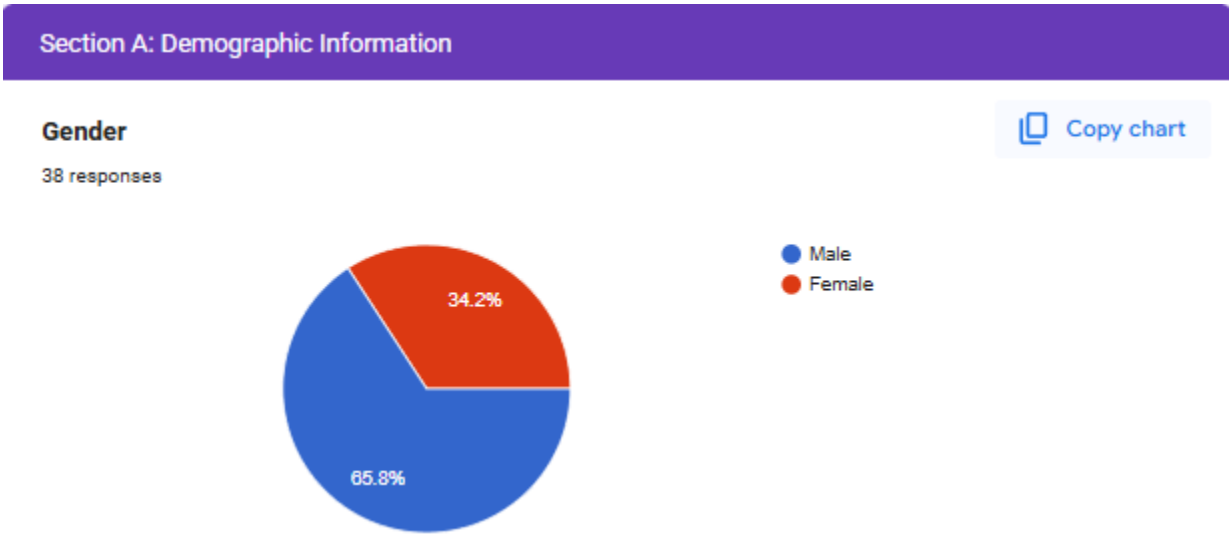


Figure 1: Gender Distribution

The Survey participants were presented with gender indicators, as shown in the diagram. The entire sample consisted of 38 people. It was revealed that 65.8% respondents were male and 34.2% were female.

The findings reveal that the majority of participants have been a male. This scenario resonates with a number of IT and cybersecurity careers, where men assume the majority of the positions held by workers and students. Nevertheless, there is a female respondent who also makes significant contributions that help provide diverse insights regarding cyber safety issues.

To sum up, the survey responses were collected through participants with diverse gender representation, making the insights a broader view of security config practices and ransomware awareness.

3.3.2 Age Group Distribution

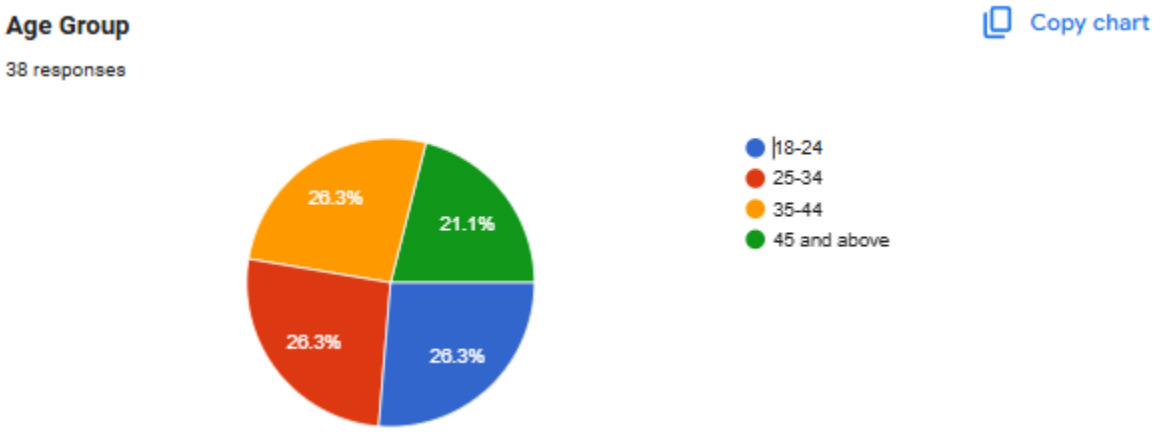


Figure 2: Age Group Distribution

The results, which indicate the age of the respondents, are shown in Figure 2. According to the survey information, respondents were found to be distributed evenly across different age groups.

The age categories of 18–24, 25–34, and 35–44 each account for 26.3% of the total respondents, while 45 or older groups account for 21.1%. The results highlight the fact that the participants in this survey were among both younger and more experienced age groups.

The representation of respondents of the age of 18 to 24 means students or early-career IT professionals got a chance to participate, while the respondents of the age of 25 to 34 and 35 to 44 belong to the group of IT industry professionals.

Moreover, professionals aged 45 and above testified to the survey's relevance, as several contributed. This stable age ratio has given an insight into information security configuration practices and ransomware awareness among professionals of different experience levels in the IT domain.

3.3.3 Current Role / Occupation

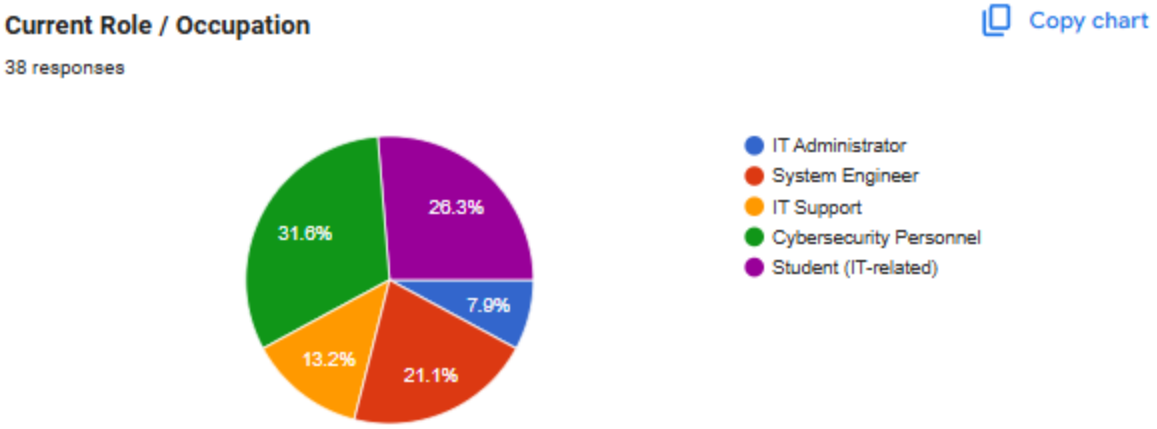


Figure 3: Current Role / Occupation

The distribution of respondents by current role (or occupation) is presented in Figure 3. The data shows that the highest number of participants was made up of cybersecurity personnel, who account for 31.6% out of all the respondents. This suggests that a significant portion of respondents have hands-on experience in cybersecurity-related activities.

Furthermore, 26.3% of the respondents were IT-related students, suggesting that students familiar with information technology and cybersecurity also joined the survey. At the same time, 21.1% of respondents were system engineers, and 13.2% worked in IT support positions, both of which are technical positions that administer and operate IT infrastructure.

7.9% of the respondents had IT administrators, which is the least important group in the survey. Despite this, hardware and software admins usually have the central role of efficient system growth and the implementation of protection guidelines.

The output confirms the survey's ability to collect answers from different cohorts of the IT profession. Professionals in cybersecurity, systems engineers, IT support staff, and students' presence is a critical factor in obtaining a balanced opinion about the security configuration measures and ransomware awareness.

3.3.4 Years of Experience in IT

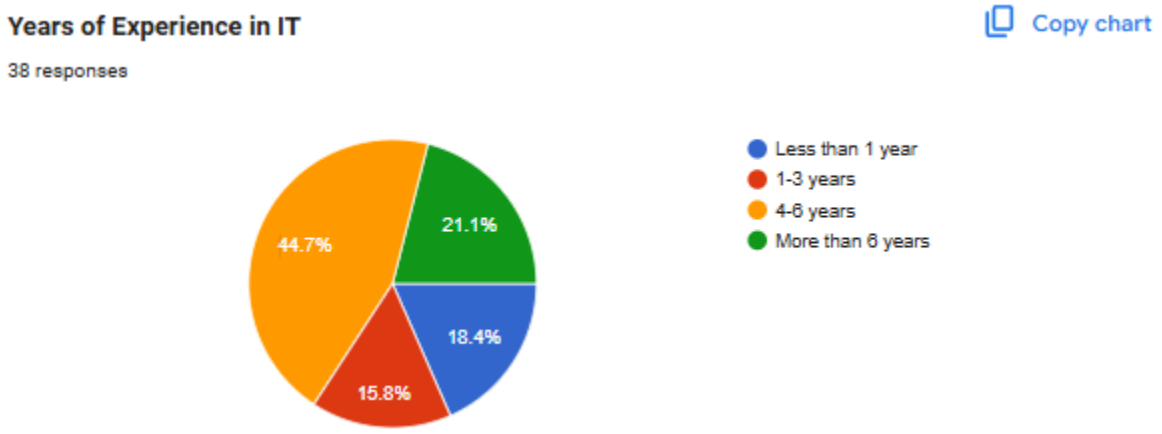


Figure 4: Years of Experience in IT

The last characteristic that we considered was the years of experience in the field. This distribution of respondents by years of experience can be seen in Figure 4.

The survey results demonstrate that the maximum number of respondents are IT professionals falling under the age of 4-6 years category, which constitutes 44.7% of the total respondents. These findings suggest that quite a number of respondents have a fair amount of professional experience and have possibly developed practical skills revolving around the management of IT systems and their security.

The 21.1% of respondents who claimed to have had more than 6 years of experience indicated that some sophists took part in the survey. These experts possibly understand better the theory of system configuration management and have a practical insight into the security issues of IT systems.

On the other hand, there were 18.4% of respondents who had less than 1 year of experience, and 15.8% had from 1 to 3 years of experience. These three groups consist of either final-years IT students or young professionals with limited working experience.

In general, the analysis showed that the respondents represent a range of experience levels, starting from beginners and going all the way up to highly qualified professionals. This heterogeneity is useful for forming a much larger picture about the practices currently adopted in security configurations and some of the major threats from ransomware that the people operating in the IT community are aware of.

Section B: Security Configuration Practices

3.3.5 Awareness of System Security Configuration Checks

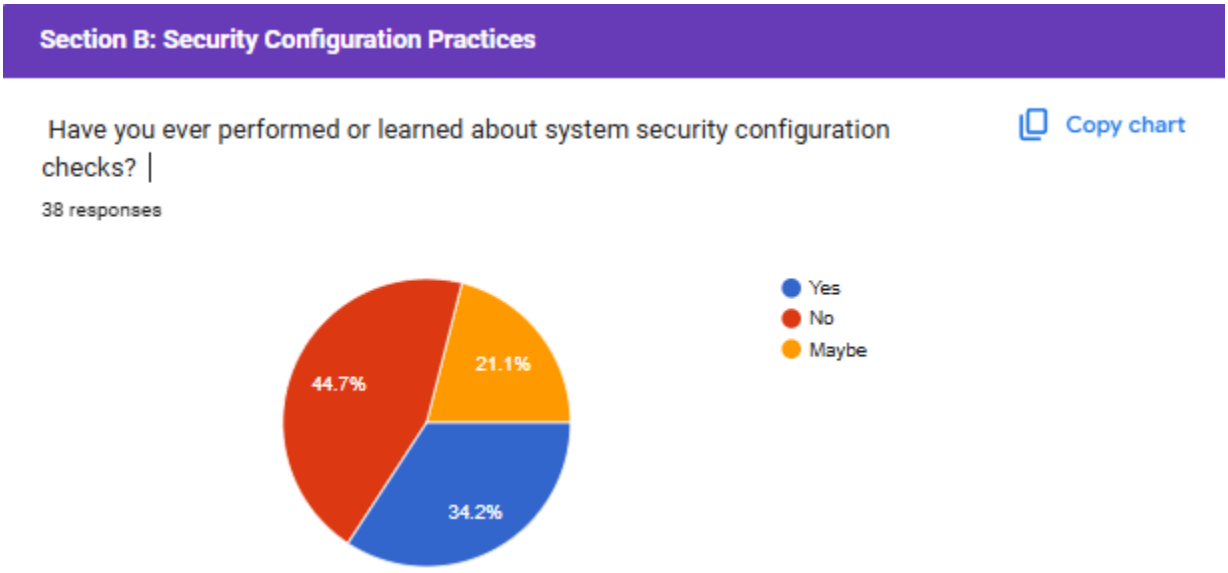


Figure 5: Awareness of System Security Configuration Checks

Figure 5 illustrates the survey findings regarding the recognition of system security configuration checks. From the selection, it is clear that 44.7% of all respondents raised that they have never had any experience or knowledge in system security configuration checks, being the highest percentage. This data shows that a sizeable part of respondents failed to acquire either practical or theoretical knowledge in the field of security configuration management.

On the other hand, 34.2% of the respondents asserted that they have performed or learned about system security configuration checks, proving that a fraction of the participants has a background in security performance and the practice of system configuration evaluation. Most likely, such respondents will be working in the role of technical support or possess the hands-on experience necessary for system administration or cybersecurity.

Besides that, 21.1% of the respondents opted for Maybe, which stands for uncertainty or minimal amusement toward system security configuration processes. The results indicate that although some of the participants knew the topic in general, they may have had no first-hand experience of going through the configuration checks.

Generally, the findings indicate that acceptance and perception of system security configuration checks may remain low among some respondents. This data confirms the relevance of the technologies that are intended to facilitate security configuration assessment and assist organizations in discovering possible vulnerabilities with ease.

3.3.6 Frequency of Reviewing System Security Configuration

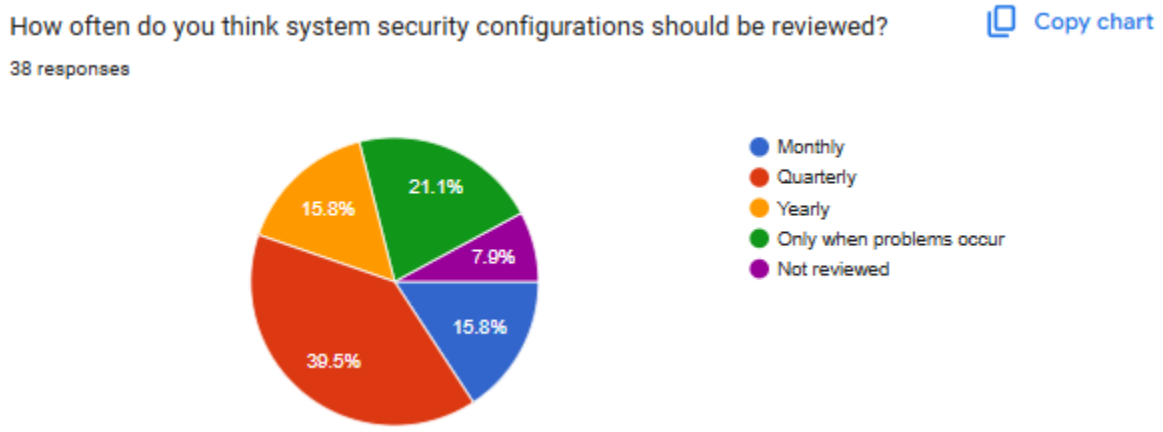


Figure 6: Frequency of Reviewing System Security Configuration

The number of reviews of system security configurations is illustrated in Figure 6. The majority of respondents, 39.5%, indicated that system security configurations should be reviewed quarterly. This implies that periodic reviews are believed to be important by a number of participants to ensure that the system configurations are secure and up to date.

Furthermore, 15.8% believed that security configurations should be reviewed on a monthly basis, which means that a part of the participants prefers more frequent monitoring in order to counteract their security risks. On the one hand, a group of 15.8% selected yearly reviews, which could reflect organizations that have limited resources or do not require that much security.

On the other hand, 21.1% of the respondents stated that system configurations should be reviewed only when problems arise, signaling that some organizations will deal with security management in a reactive way, not in a proactive way. A conspicuous 7.9% of respondents indicated that configurations are not reviewed, which could pinpoint a lack of attention to security.

The results indicate that the majority of respondents, although acknowledging the value of configuration reviews, do not have a common understanding regarding the exact frequency. The finding furthermore calls for the implementation of automatic configuration tools for organizations to periodically monitor their security level in an efficient way.

3.3.7 Familiarity with Centralized Security Configuration Scanning Tools

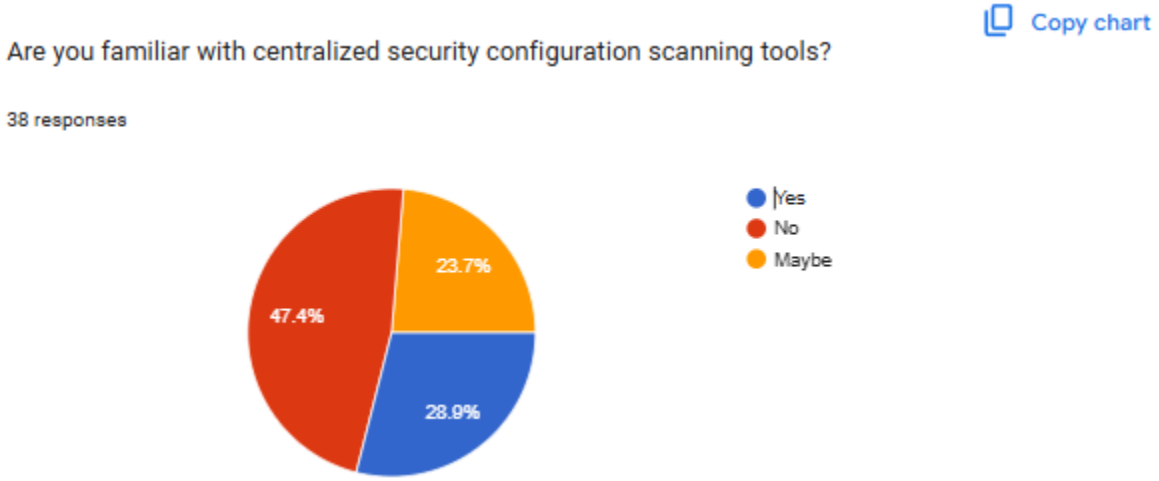


Figure 7: Familiarity with Centralized Security Configuration Scanning Tools

Figure 7 presents the survey results for the question about knowledge and experience with centralized security configuration scanning tools. The results indicate that 47.4% of respondents did not specify being familiar with such tools, belonging to the largest group's category. This may identify that many individuals are not fully exposed to centralized security configuration scanning solutions in the present work or study environments.

At the same time, 28.9% of respondents stated being familiar with centralized configuration scanning tools, which points out that some participants have either work experience or knowledge related to automated security assessment tools. The observation applies to personnel performing the technical roles of cybersecurity specialists, system engineers, or IT administrators.

Furthermore, 23.7% of respondents selected “Maybe,” which means they are not fully confident in the concept or application of centralized configuration scanning tools. It may indicate that some participants demonstrate indirect exposure to such tools but may not be entirely informed about their exact features.

In conclusion, the results demonstrate that many respondents still have a limited understanding of centralized security configuration scanning tools. This finding further supports risk reduction through a user-friendly and centralized platform that allows organizations to memorize, process, and analyze incoming information easily.

3.3.8 Experience with Ransomware Attacks

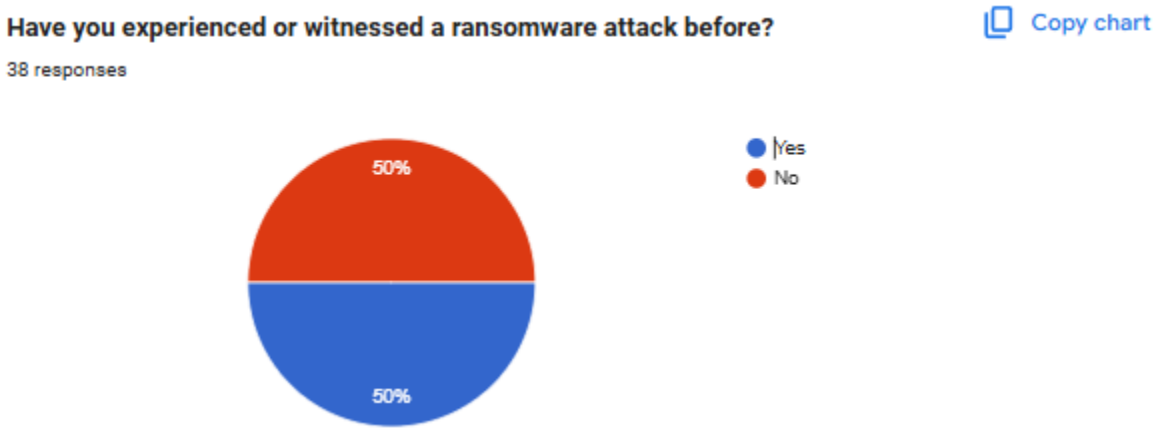


Figure 8: Experience with Ransomware Attacks

Ransomware attack experience survey displays results according to Figure 8. The outcomes suggest that 50% of the respondents had in the past experienced or seen a ransomware attack. The other 50% said that they had not come across such occurrences.

Since responses seem to be evenly split, we can infer that ransomware is a relatively common threat, with a large number of participants having directly encountered incidents or witnessed them in their respective environments. While experiencing ransomware attacks personally, respondents may have become aware of the negative effects of insecure configurations.

However, respondents who have not had the misfortune of experiencing ransomware attacks may be notified of such risks through training, schooling, or even knowledge associated with the industry. This shines a light on how a case study can be beneficial in driving home the need for secure put in place practices.

All in all, the data displayed indicates that cryptocurrency services still remain an issue in cybersecurity. The conclusions also highlight the importance of designing tools that will help companies to find security misconfigurations and consequently decrease the risk of infection with ransomware.

3.3.9 Common Causes of Ransomware Infection

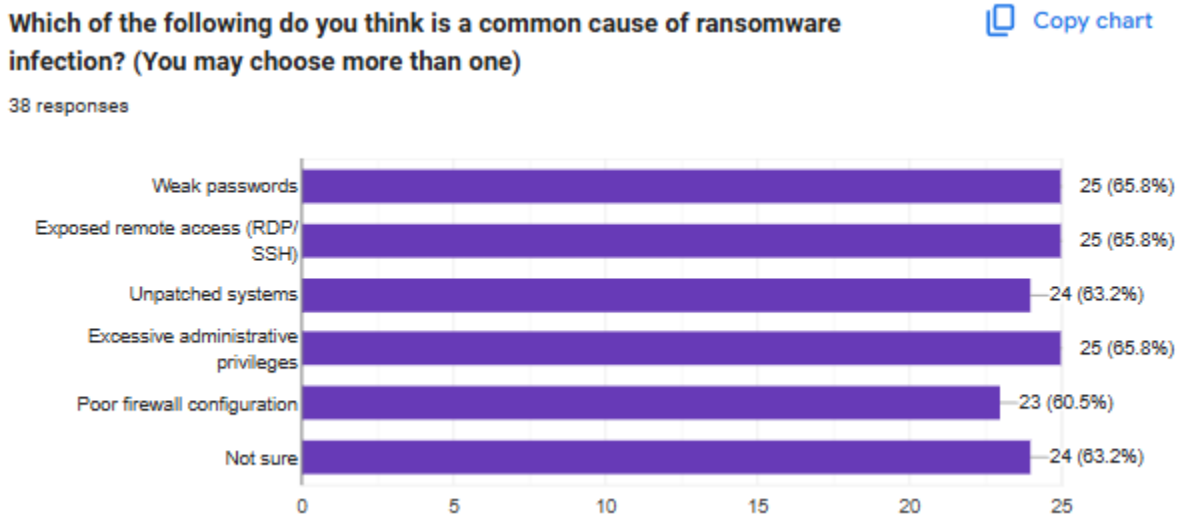


Figure 9: Common Causes of Ransomware Infection

The diagram displayed in the following paragraph represents the considered potential causes of ransomware infections. It must be remembered that the respondents could choose more than one option during the survey. Thus, percentages are the portion of survey participants that selected each of the items.

Weak passwords, exposed remote access services (for instance, RDP or SSH), and an excess of administrative privileges were highlighted as the alleged reasons of ransomware infections by 65.8% of the survey respondents, thereby being the top three. These vulnerabilities are broadly considered as security issues allowing intruders to gain unauthorized system access.

Unpatched systems, with 63.2%, were another common factor noticed among the survey participants, signifying that many respondents are aware of the problems caused by outdated software and missing security patches. Likewise, 60.5% argued that bad firewall configuration may be a maximizer of the risk of ransomware infection, proving, again, the relevance of appropriate network security controls.

Moreover, 63.2% of the participants also chose "Not sure." This can imply that some people's views regarding the causes of ransomware attacks can still be vague. Such a result may be attributed to a broad variety of cybersecurity awareness among the respondents.

As an overarching conclusion, the survey indicates that the respondents have an overview of some technical measures that may lead to ransomware infections, notably weak authentication mechanisms and exposed remote services. These results, speaking for the necessity, demonstrate the utility of automated detection of security misconfigurations and its role in lowering the risk of ransomware attacks.

3.3.10 Perception of the Relationship Between System Misconfiguration and Ransomware Risk

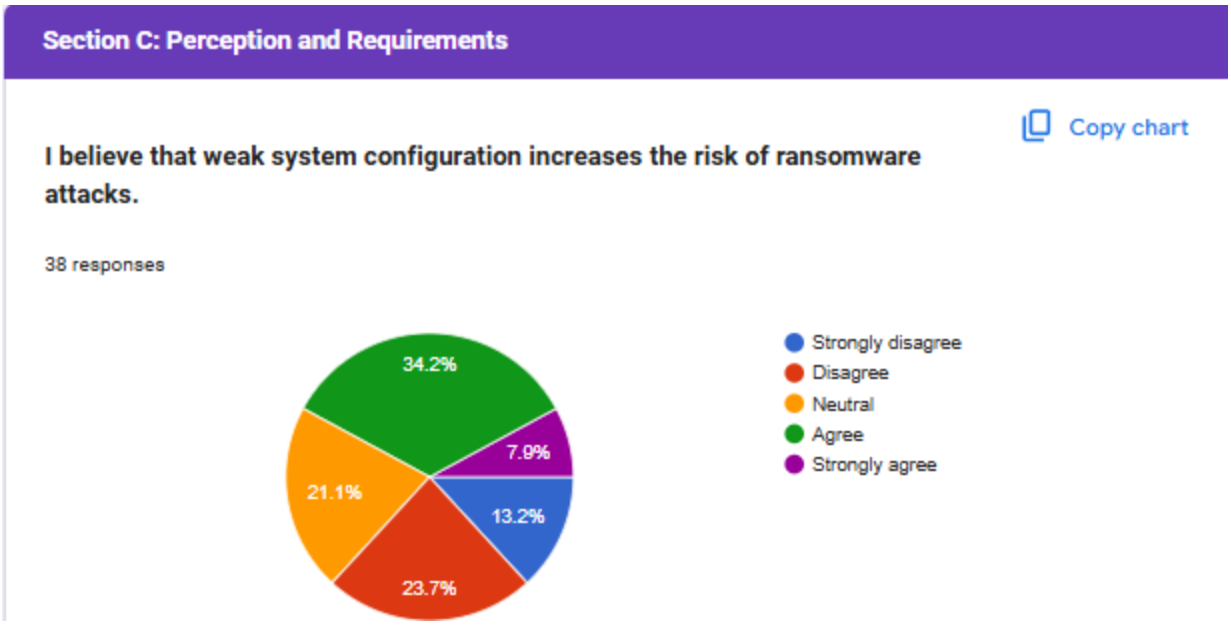


Figure 10: Perception of the Relationship Between System Misconfiguration and Ransomware Risk

Figure 10 displays the survey statistics about the percentage of reactions on the link between weak system configurations and ransomware attacks. The Likert scale was used to record respondents' answers, from strongly disagree to strongly agree.

Only 34.2% of respondents agreed that weak system configuration increases the risk of ransomware attacks, whereas only 7.9% strongly agreed with the statement. This translates to only a part of respondents being cognizant of the risk of ransomware threats by misconfigured systems.

Around 23.7% of those interviewed did not agree with the statement; 13.2% strongly disagreed, which shows that in some cases the subjects studied find it difficult to realize the part that system misconfiguration has in facilitating ransomware attacks. To caution you, out of the total reached respondents, 21.1% chose to be neutral for this matter, perhaps to indicate their uncertainty or lack of interest in it.

Thus, it appears that only a minority of respondents are concerned about this issue related to weak system configurations, but the majority disagrees or doubts. This implies that there is a need to educate managers and equip them with the right tools that can swiftly find bad configurations that lead to ransomware threats.

3.3.11 Perception of the Effectiveness of a Centralized Configuration Scanning Tool

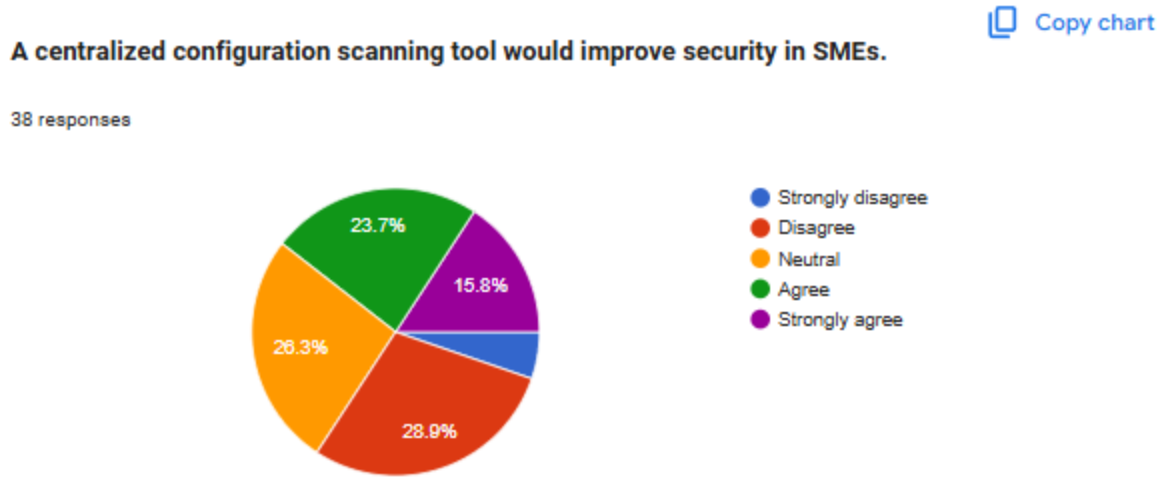


Figure 11: Perception of the Effectiveness of a Centralized Configuration Scanning Tool

The figure reveals the responses of a survey regarding whether a central scanning tool's configuration would bolster the security of SMEs. A five-point Likert scale, ranging from strongly agree to strongly disagree, was used for the survey.

From the result, 23.7% agreed and 15.8% strongly agreed that the security of SMEs could be improved with a centralized scanning tool. The answers of respondents imply that some of them expect organizations to introduce automated tools to detect misconfigurations and prevent subsequent security incidents to be wise.

Disapproval from 28.9% of respondents with only 4.3% strongly disagreeing can be observed. Moreover, 26.3% of participants took a neutral stance, suggesting that anonymity provided them a platform to state their thoughts on centralized monitoring tools.

The latter positions in this case represent two opposing aspects of the same utility of centralized configuration scanning systems. In fact, the number of respondents being oriented towards the aforementioned system can be justified as proof that a tool should be specifically developed for SMEs to facilitate the misconfiguration detection and ensure the improvement of their cybersecurity threats.

3.3.12 Preferred Features of a Ransomware-Focused Configuration Scanner

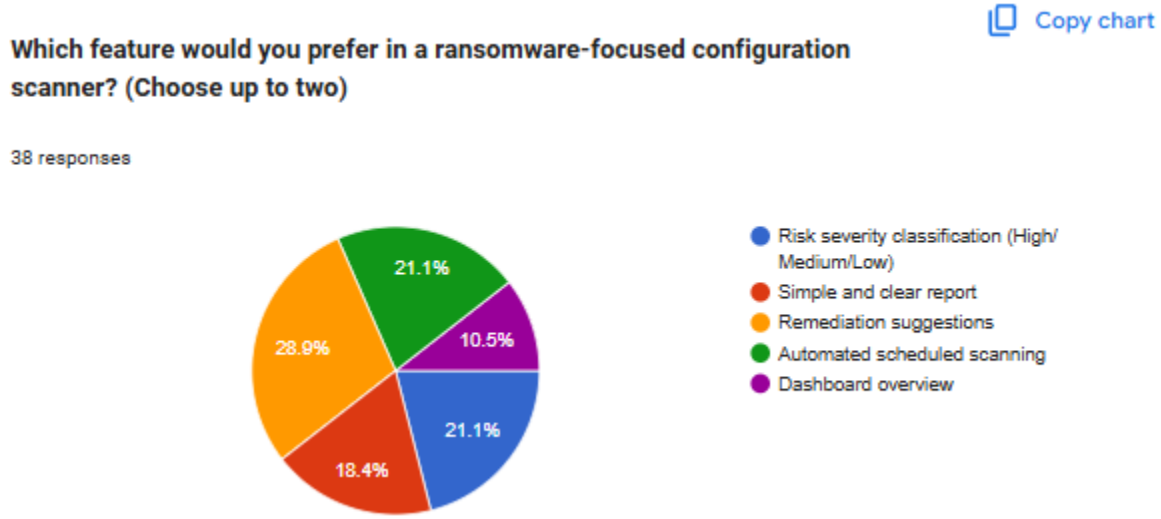


Figure 12: Preferred Features of a Ransomware-Focused Configuration Scanner

The findings obtained from the investigation where users gave their opinions about the features of the configuration scanning for ransomware are listed in the figure below. Percentage is used here, showing the part of the total number of respondents who checked that feature.

The results show that 28.9% of the respondents preferred remediation recommendation feature the most. This points out that many users also expect that, besides finding security vulnerabilities, these tools should have some feature for providing suggestions regarding their correction.

Moreover, as many as 21.1% of the respondents showed the risk severity classification (High/Medium/Low) and automated scheduled scanning as their preferred options. The first functions will help admins in assessing the security alert level, while the last one will enable them to conduct automatic system security checks.

On the flip side, an indicative 18.4% of the people exhibited liking for summarized, comprehensible reports, which means that both aspects of user-friendliness and ease of communication of results were given equal importance by the end-users. After all, there are only 10.5% of the respondents who favor a dashboard, which further supports the idea that many users prefer to get consolidated security info.

To conclude, users look for some useful functions that will help them to detect security problems and mitigate these issues promptly. This is what the outcomes imply, and the findings supply a convenient basis for defining desired functionality for the application under consideration.

Chapter 4

4.1 Introduction

The design and implementation of the proposed centralized security configuration scanning tool is discussed in this chapter. The requirements collected during the investigation phase were used to design the system to have a centralized platform for security configuration assessment of both Linux and Windows operating systems. The design aims to assist with the automation of risk assessment and ransomware detection of security misconfigurations in a way that allows for easier detection by administrators.

The chapter starts by introducing the overall system architecture and workflow, and then provides the system design diagrams such as architecture diagram, use case diagram, sequence diagram, risk scoring process diagram, pseudocode diagrams, and entity-relationship diagram (ERD). The diagrams show how various parts of the system work with each other and how the scanning is done.

Then the interface design of the system is presented, showing how the system's functionality is offered to administrator using the web-based dashboard. Lastly, the implementation of each of the main modules is discussed, such as the Flask dashboard, Linux scanner, Windows scanner, risk scoring engine, scheduled scanning module, and PDF report generation. When combined, they provide a single solution to help organizations, especially SMEs, assess their security settings and minimize the risk of ransomware attacks.

4.2 Design

4.2.1 Architecture Diagram

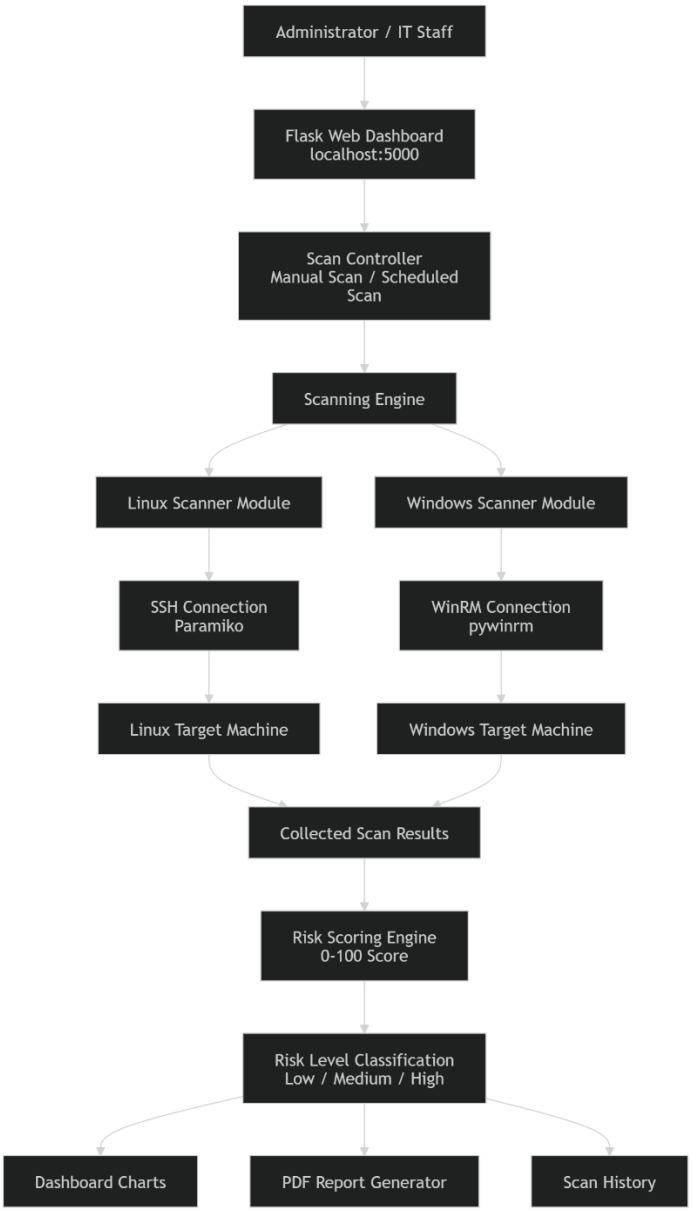


Figure 13: Architecture Diagram

The proposed system is based on the centralized client-server architecture. The Flask web application serves as a central management point for users to configure target machines, run security scans, view the scan results, and generate assessment reports. All the scanning is managed from a single dashboard and doesn't need any manual checks on individual devices, thus making the process more efficient.

If a scan is started, the system identifies the operating system of the machine that was selected. In the Linux case, the scanning module creates a secure SSH connection to gather security configuration information with the Paramiko library. For Windows systems, the scanner interfaces via the Windows Remote Management (WinRM) protocol, via the pywinrm library. The security information gathered is then passed to the risk scoring engine for analysis.

The risk scoring engine applies to each security configuration a set of pre-defined security rules, and it returns an overall risk score. The results of the scan, risk level, and timestampts and remediation recommendations are saved to the database for later reference. Lastly, the administrator can use the dashboard to check the assessment or download the results of the assessment in PDF format to stay docu.

This centralized architecture is more efficient in the system management and enables the administrator to conduct security evaluations from a single platform on multiple machines.

4.2.2 System Workflow

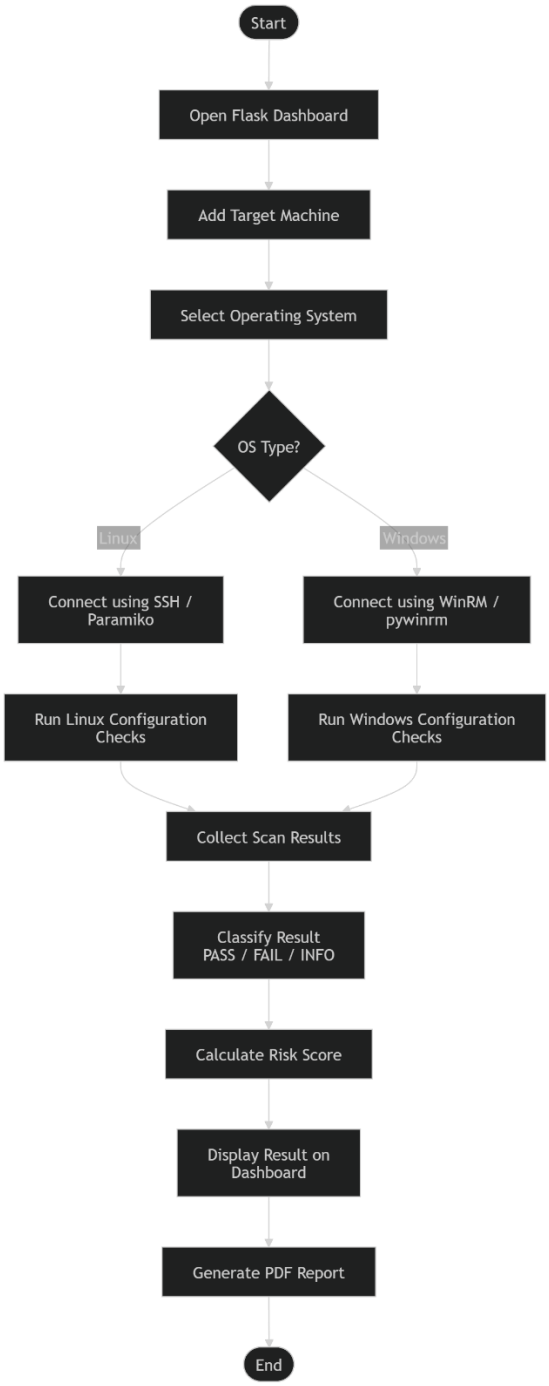


Figure 14: System Workflow

The proposed system overall workflow is shown in figure 14. It starts from when an administrator logs into the dashboard and chooses a registered target machine. Once the target has

been selected, the system detects whether it is running a Linux or Windows OS before starting the scanning module for the respective operating system.

The chosen scanning module creates a secure remote link to the targeted machine and checks for a number of pre-defined security configuration checks. The collected results are then sent to the risk scoring engine, which then assesses each configuration it detects and assigns an overall security risk score.

After an assessment, the scan result is saved in the database and will be shown on the dashboard. Administrators can view detected security issues, check recommended remediation actions and create a PDF report for documentation. The process of assessing can be fully automated with hardly any human effort.

4.2.3 Use Case Diagram

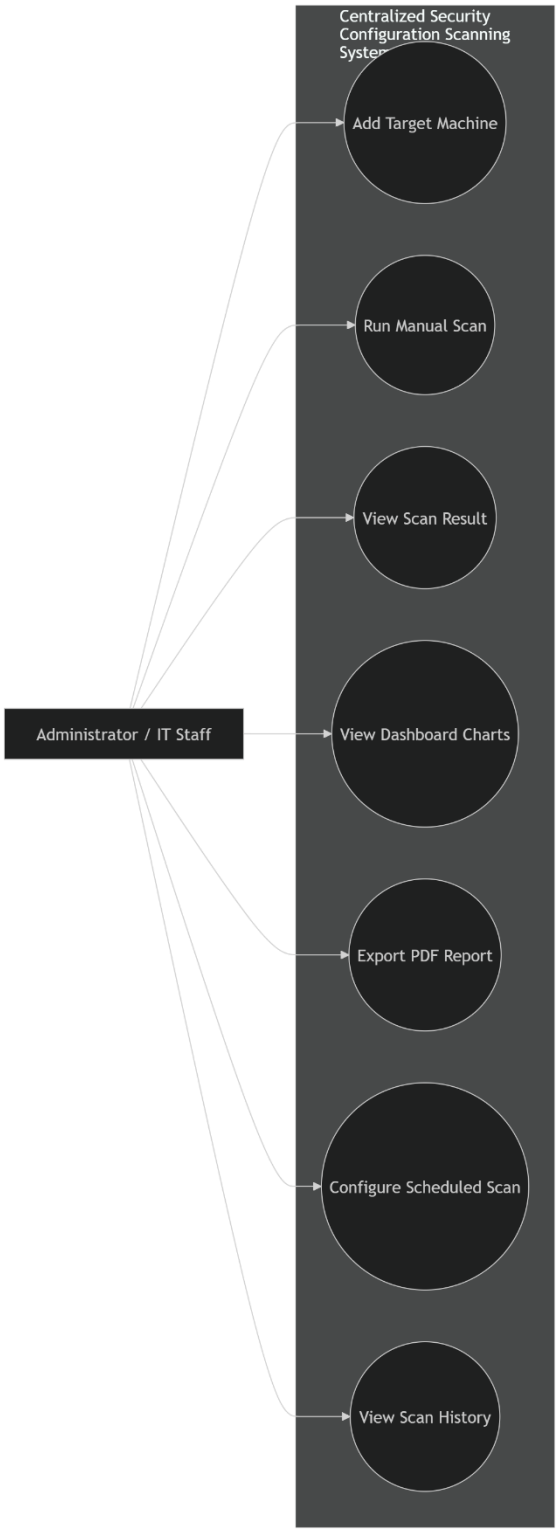


Figure 15: Use Case Diagram

The Use Case Diagram represents how the administrator will interact with the proposed centralized security configuration scanning system. The administrator is the main user in this project who will be responsible for the scanning process and will be able to use all of the functionalities of the system via a web-based dashboard. The overall system is easier to manage and operate as only one user role is required as the system is mainly designed for SMEs.

The administrator must register target machines before assessing can take place. The administrator will input the target machine information, such as IP address, operating system type, user and authentication information, during the process. They are securely stored and accessed when manual or scheduled scans are triggered.

After adding the target machine, the administrator can start a security configuration scan. It detects the type of machine (Linux or Windows) and enables the appropriate scanning module. During the scanning process the administrator does not have to review each configuration as the system checks it automatically in the background.

After the scan, the administrator can view the scan results on the dashboard. The results consist of security misconfigurations found, risk scores calculated, overall risk and recommended remediation measures. The results of this analysis provide information about the state of security on each machine and enable administrators to determine the priority of corrective measure according to the severity of the problems found.

Lastly, the administrator will be able to create a PDF document report for documentation and future reference. All the required assessment information such as target details, scan results, timestamps, calculated risk scores, remediation suggestions, etc. are included in the generated report. This feature makes auditing more easy and allows the organization to keep security assessment history.

From the overall perspective, the Use Case Diagram shows that the proposed system allows for a seamless flow of actions, enabling the administrators to perform various security assessment tasks in an efficient manner and within a single platform.

4.2.4 Sequence Diagram

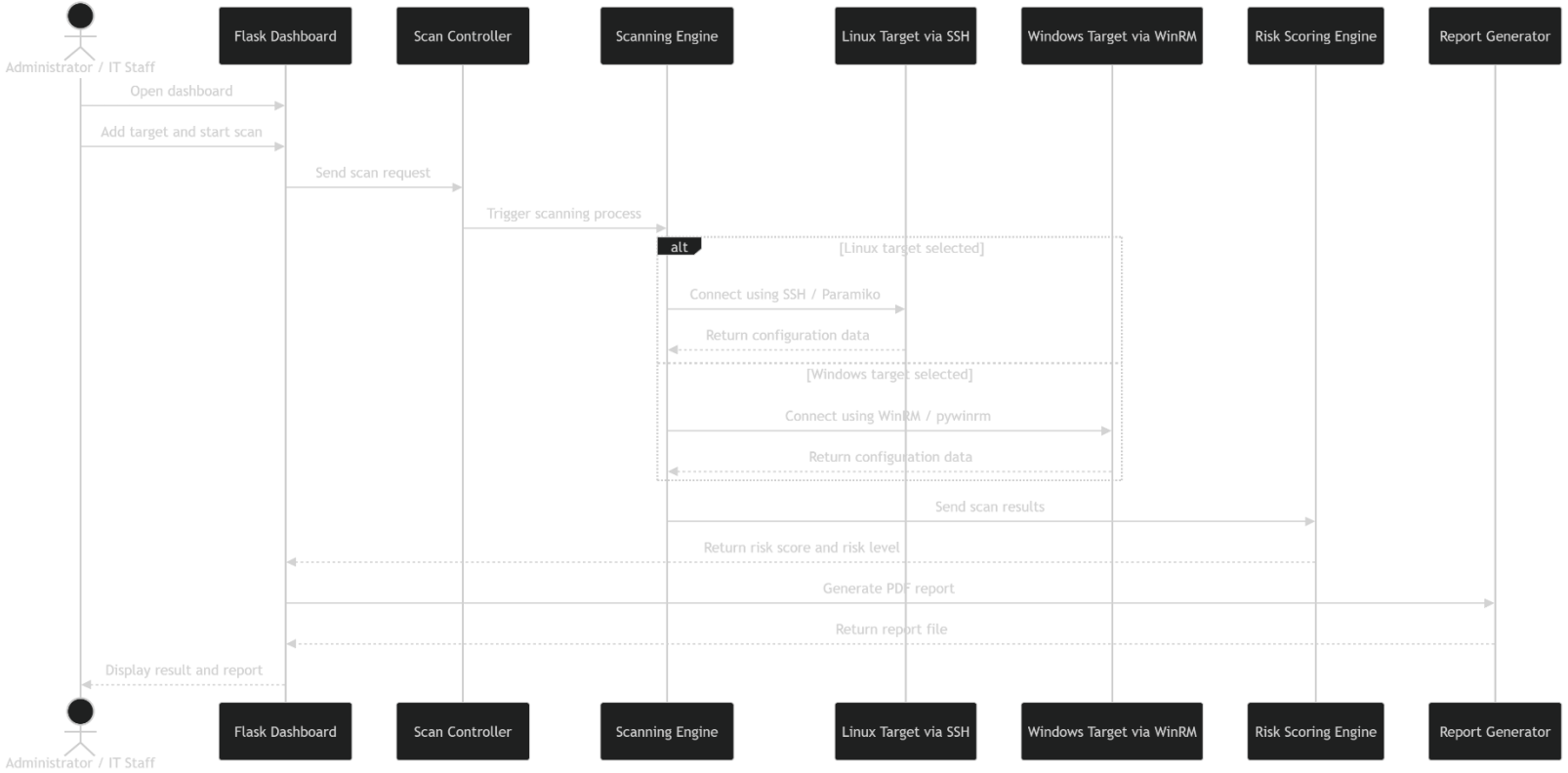


Figure 16: Sequence Diagram

The Sequence Diagram shows the sequence of interactions between the administrator, the Flask web app, the scanning modules, the risk scoring engine, the database, and the report generation module when conducting a security assessment. It offers a time-line of the flow of information into the system from the start of a scan to the end and the creation of the final report.

It starts with the administrator choosing a registered target machine from the administrator dashboard and requesting a scan. The Flask application accepts the request and first checks in the database the targeted data. Based upon the operating system selected for the target machine, the system deems whether to run the Linux scanning module or the Windows scanning module.

The chosen scanning module creates a secure remote connection, either through SSH or WinRM, and checks various security settings. At this point, information regarding the status of the firewall, the firewall's password policy, its remote access configuration, whether software is up-to-date, whether an administrator account is enabled, and other firewall security settings gets collected. The scanning process is finished and the data collected is sent back to the Flask application.

The Flask application passes the results of the scan to the risk scoring engine where each security issue is scored based on predetermined criteria. Depending on the calculated score the target machine is placed in one of three risk categories: Low, Medium or High. Once the evaluation has been made, all the assessment data is saved in the database.

Finally, the stored information is retrieved and displayed to the administrator on the dashboard with the scan results. The administrator may create a PDF report if needed, with the assessment results and recommendations for remediation. The sequence diagram shows that every component has a specific responsibility, and the whole assessment process is modular, structured and easy to maintain.

4.2.5 Risk Scoring Diagram

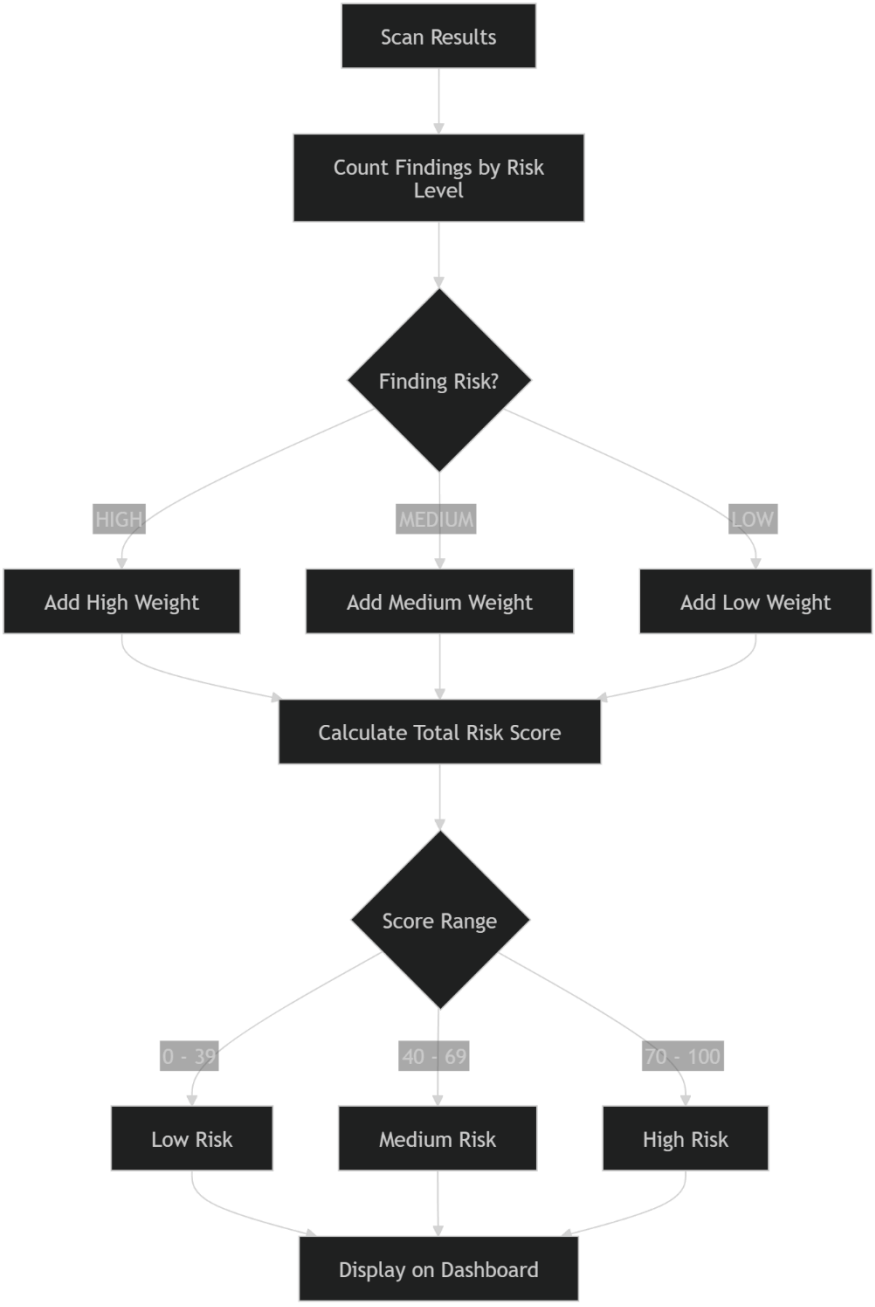


Figure 17: Risk Scoring Diagram

The Risk Scoring Diagram shows the risk score of a target machine based on its overall security posture once the security machine configuration assessment has been performed. The

system doesn't just report a single scan result, but rather aggregates all the security problems that it has identified into a numerical risk score, allowing administrators to get a better picture of the security situation in the system.

The risk evaluation process starts once after the scanning module has performed all pre-defined security configuration checks. Everyone configuration found is assigned a severity level based on the impact on system security. Findings that pose a higher risk of ransomware attacks, such as: Disabled firewalls, Weak password policy, Exposed remote services, and Outdated software add more risk points than less critical findings.

All detected security issues are evaluated and then a sum of the scores is calculated. A final score of 0-100 is given and the score is known to be in one of three predetermined ranges. A score of 0-39 is considered Low Risk, where minor security issues have been identified. A score of 40-69 is considered a Medium Risk and indicates that multiple security vulnerabilities need to be mitigated, as there is a potential for further security issues. A score of 70 - 100 is considered High Risk, meaning the target machine has several critical security misconfigurations that need to be addressed.

The system not only shows overall risk level, but also gives remediation tips for each problem identified. These recommendations help administrators determine the proper corrective actions to be taken without having extensive cybersecurity knowledge. The proposed system integrates automated scanning with a user-friendly risk classification system that empowers SMEs to focus their security efforts more effectively and minimize the threat of ransomware attacks.

4.2.6 Linux Scanner Pseudocode Diagram

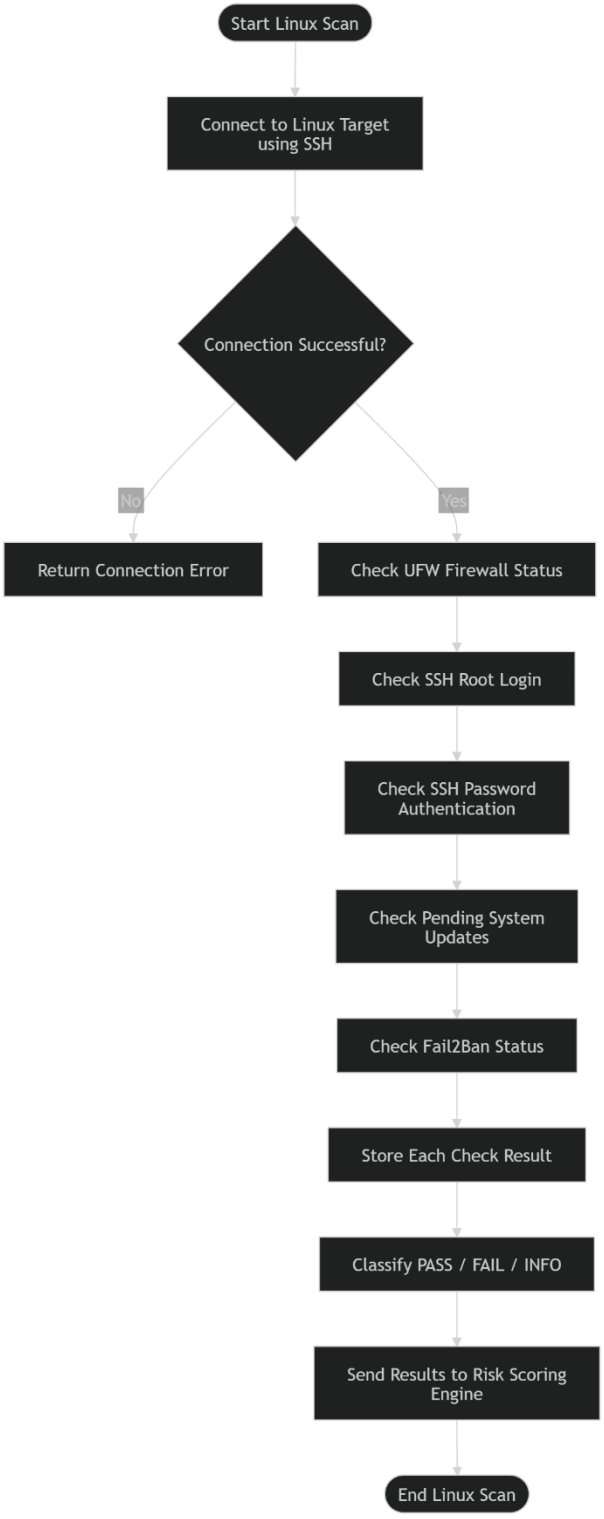


Figure 18: Linux Scanner Pseudocode Diagram

This is a pseudocode diagram for the Linux Scanner.The Linux Scanner pseudocode diagram.The Linux scanner Pseudocode Diagram shows the order of logical operations that the Linux scanner module uses to determine the security configuration of a target machine. The pseudocode does not elaborate on the implementation of the program code, but it explains the general steps the system takes throughout the scanning process.

The initial step in the scanning is to connect to the target Linux machine using a secure Secure Shell (SSH) connection created between the centralized scanning system and the target machine with the help of the Paramiko library. The system first checks the connection credentials provided are valid before it performs any security checks. If authentication cannot be completed, or the machine to be scanned cannot be found, the system stops the scanning session immediately and returns an appropriate error message to the administrator. With this validation step, unnecessary attempts of scanning are avoided and the reliability of the system is enhanced.

Upon successful connection, the scanner checks a series of preset security configuration checks. These include checking if the Linux firewall is enabled, if SSH root login has been disabled, if the SSH password authentication is in line with secure configuration practices, if the operating system has the latest security patches, and if Fail2Ban is installed as a means to protect the system from brute force attacks. The following security configurations were chosen as those which are tightly connected with the ransomware prevention and are generally prescribed by cybersecurity best practices.

Every security check is written as PASS, FAIL or INFO. If the result is PASS, the setting is considered to be within the recommended security configuration, but if the result is FAIL, there is a security issue that must be addressed. INFO is used to supplement information that could help administrators while reviewing without impacting the overall risk score.

Once all the security checks have been verified, the Linux scanning module collates all of the findings and passes them to the risk scoring module for further analysis. This is then wrapped up safely to prevent an open connection to the target machine. This workflow guarantees a uniform, secure and efficient approach to each security assessment on Linux, with minimal manual involvement.

4.2.7 Windows Scanner Pseudocode Diagram

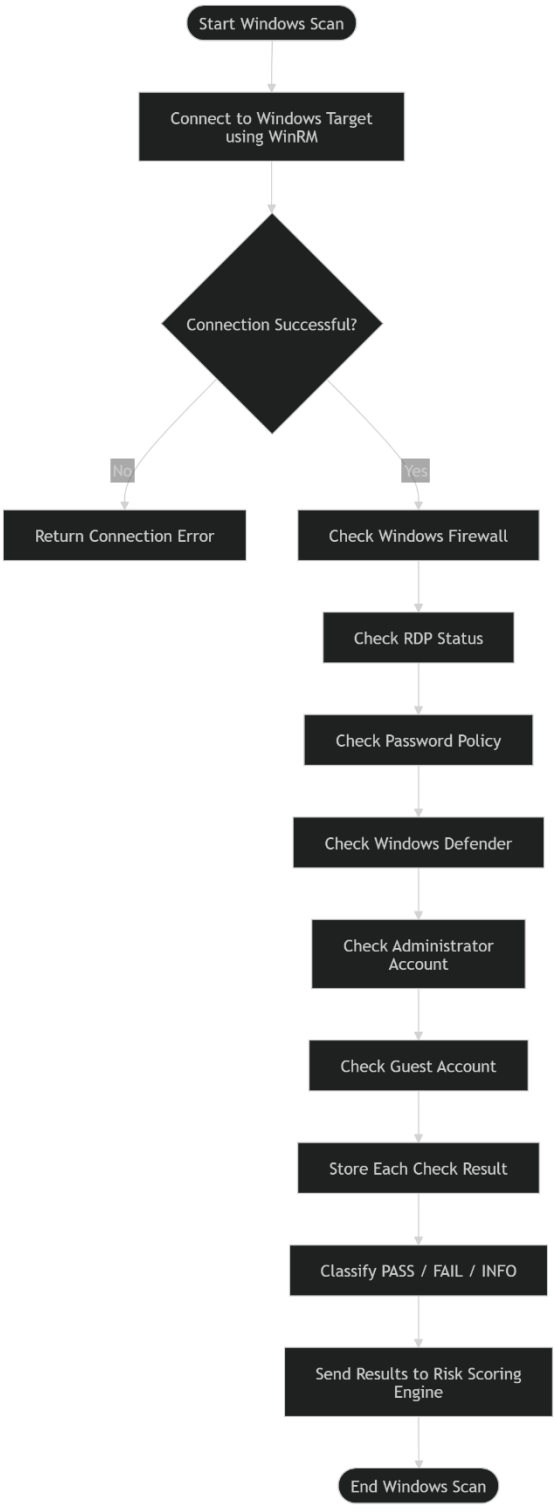


Figure 19: Windows Scanner Pseudocode Diagram

The following is the Windows Scanner Pseudocode Diagram that explains the order of operations performed by the Windows scanning module in the security configuration assessment. Like the Linux scanner, the pseudo code shows a simplified logic of the system without concentrating on the syntax of the code, so that readers can understand how the assessment process is made.

The scanning protocol starts as soon as the system connects remotely to the target Windows machine via the Windows Remote Management (WinRM) protocol via pywinrm. The system checks the authentication credentials passed to it at the start of assessment, making sure that the administration has the required privileges to perform the required security checks. If authentication fails or communication is not established, the scanning will halt and a message will be displayed on the dashboard.

Once connected successfully, the scanner assesses a number of Windows security settings that are often related to ransomware attacks. The assessment will check to see if Windows Firewall is enabled, if Remote Desktop Protocol (RDP) is enabled, to see what the password policy settings are, whether Microsoft Defender is enabled or not, and to see the status of the Administrator and Guest accounts.

The security requirements are defined and compared to each configuration. The results of all assessments are noted as PASS, FAIL, or INFO. The system also generates remediation suggestions for the security vulnerabilities that are discovered, which will be outlined when the scan results are presented. These recommendations help administrators make the necessary fixes without needing to have a lot of cybersecurity knowledge.

Once security checks are completed, the scanning results are then sent to the centralized risk scoring engine for further analysis. Lastly, the WinRM session is closed to prevent any unnecessary remote sessions from being opened. The structured scanning workflow enables Windows systems to be consistently assessed, and secure communication throughout the scanning process.

4.2.8 Scheduler Pseudocode Diagram

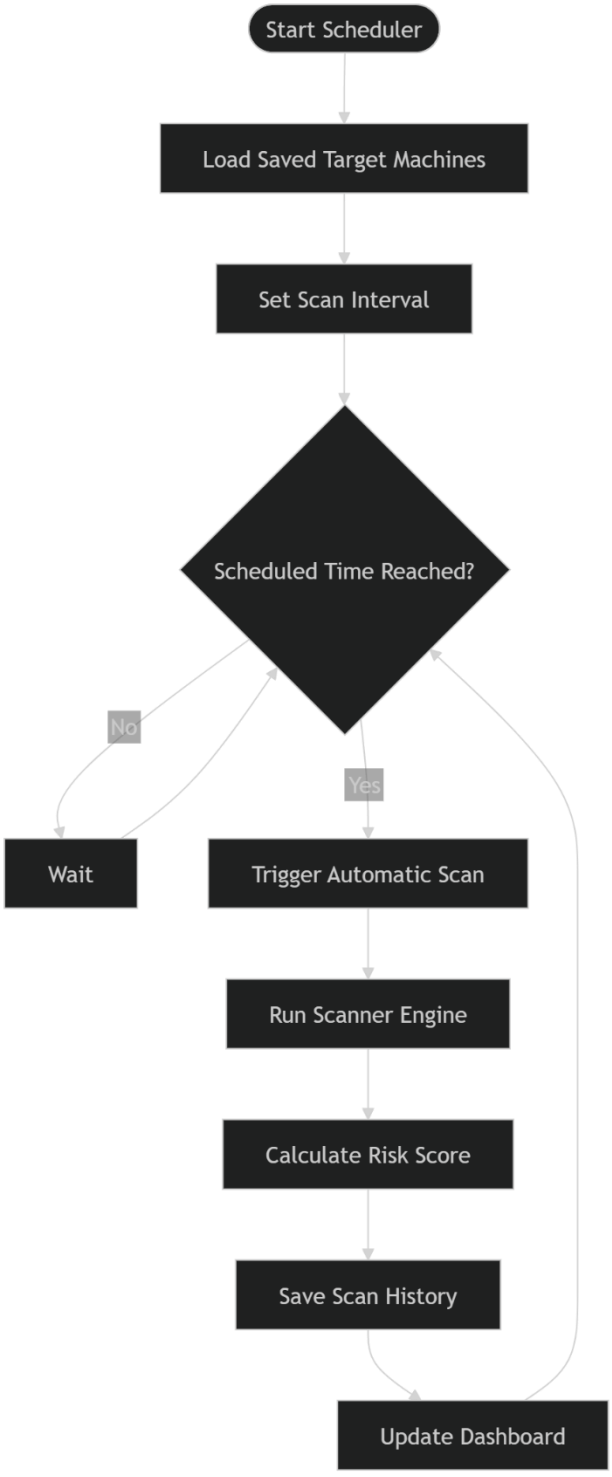


Figure 20: Scheduler Pseudocode Diagram

The automated scheduling mechanism in the proposed system is shown in the Scheduler Pseudocode Diagram. The aim of this module is to enable security configuration assessments to be performed automatically on a regular basis, without the need for constant manual action by administrators.

The scheduling process starts by an administrator setting the desired schedule of scanning in the dashboard. Scans can be configured to run every day, week, or whatever the organisation's requirements are. After the schedule is saved, the scheduler constantly checks the system clock to see if the time specified in the schedule has elapsed.

Once the time slot is reached, the scheduler automatically fetches the list of target machines registered in the database and starts scanning. Like manual scanning, the system identifies the OS of the target machine first, and then activates the Linux scanning module or the Windows scanning module. The scans are run on each target machine separately to prevent a failure in one scan from disrupting the remainder of the scans.

Once each time scheduled scan is done, the results are added to the database along with the time of scan and the risk score calculated. PDF reports have been created for future reference and compliance documents if report generation has been turned on. The administrator will then be able to view these reports on the dashboard without needing to carry out the assessment again manually.

The scheduling module has several operational benefits. Automated evaluations lower the risks of human error, make sure that security setups are evaluated on a regular basis and increase the likelihood of identifying new misconfigurations before they may be made use of by ransomware assailants. For SMEs with possibly fewer cybersecurity staff to perform routine security inspections, this feature can be especially helpful.

4.2.9 ERD

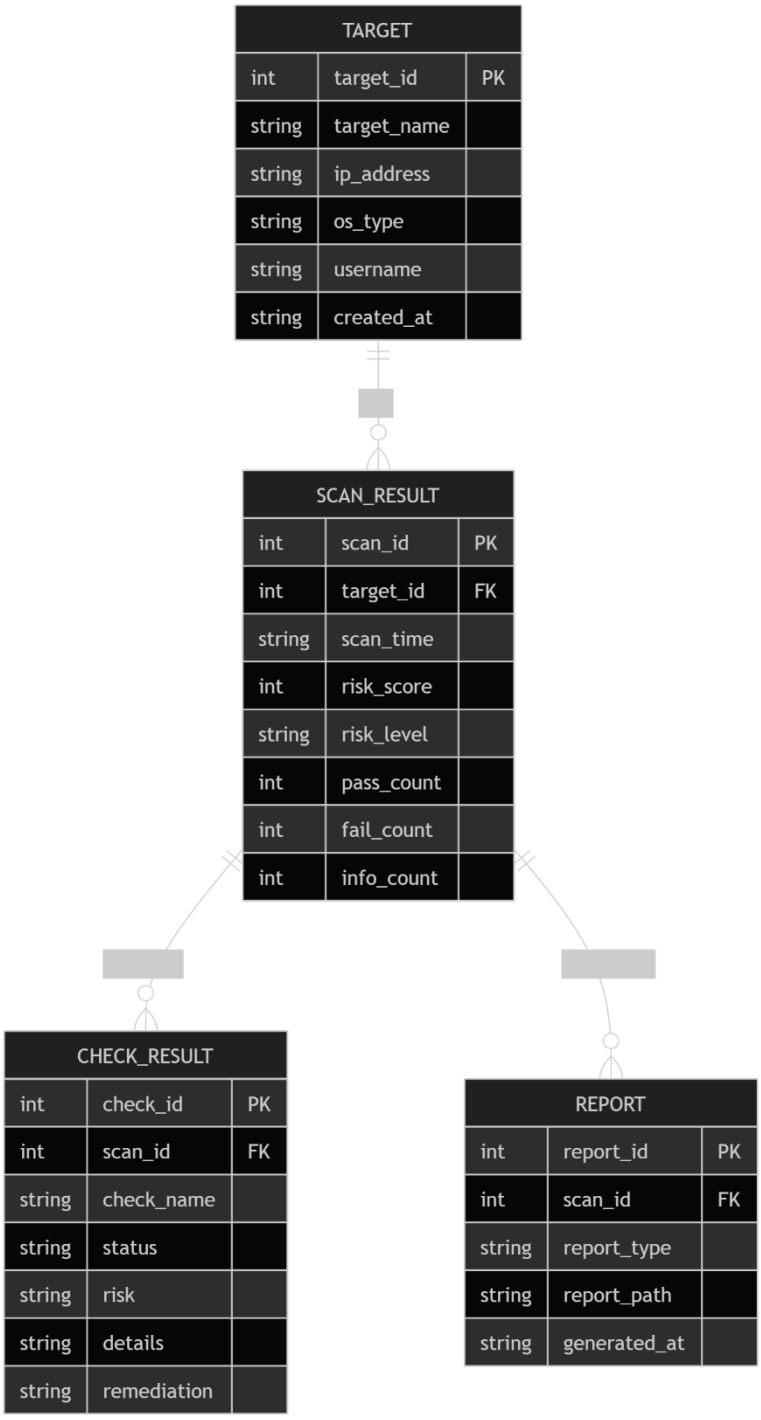


Figure 21: ERD

This is a picture of the Entity Relationship Diagram (ERD) used to design the database for the centralized security configuration scanning system. All target machine information, scan results, risk assessment information, scheduled scans and reports created by the system are stored in the central place called the database. The retrieved information can be organized in the system, and then it can be stored as related information, so that the historical assessment records can be retrieved at the same time, and the data can be synchronized at the same time.

The Target Machine entity is one of the main elements in the database that holds the data necessary for the identification and access of each registered system. The information recorded consists of: Target IP address, Operating System type, username, authentication information, and registration details. Each target machine as parent records directs the processes of subsequent scanning activities.

Scan Result is the entity which saves the result from all security assessments that are carried out on a registered machine. Each scan result is associated with a specific target machine, and contains information about the date of the scan, the specific findings that were detected, the risk score assigned to those findings, the overall risk score, and recommended steps to remediate the target machine. A one-to-many relationship exists between the Target Machine and Scan Result entities as there might be multiple assessments conducted on the same machine over time.

There's also an entity called Report, where the PDF assessment report and its metadata are stored, created by the database. Reports are connected to the corresponding scan results for future security audits or compliance reviews, and for obtaining historical documentation. There may also be other scheduling information added, for purposes of enabling automation of the scanning process and keeping track of ongoing assessment activities.

The relational database design structure reduces redundant data and makes it easier to ensure data integrity using primary and foreign keys. The design enables easy retrieval of historical scan records, future scalability and provides structure to reporting and risk analysis. The ERD provides a systematic organization of all the assessment information, so that the proposed system is reliable to run as the number of machines to be monitored grows.

4.3 Interface Design

4.3.1 Dashboard Interface

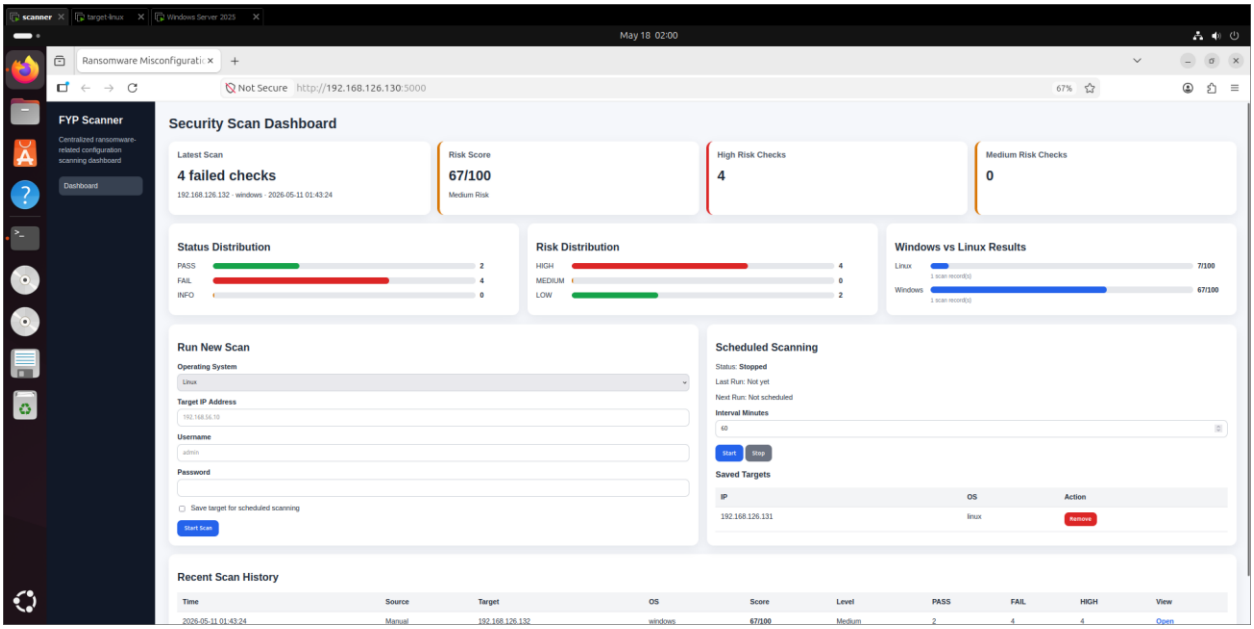


Figure 22: Dashboard Interface

This proposed centralized security configuration scanning system will have a Dashboard Interface as its control panel. It is the first page that is accessed once the administrator has successfully logged into the Web application and is the main site to perform all security assessment activities. The dashboard is designed to be easy to use and understand, allowing administrators to navigate easy the various functions without having to be particularly skilled technically.

All the target machines registered are shown in the interface along with the security assessment information of these machines. The dashboard collates all the necessary information, so that you don't have to log in to each machine. This centralized management proves to be a great increase in operational efficiency, especially for SMEs that operate a number of devices and do not have many cybersecurity staff.

It has several key performance metrics displayed on the dashboard to provide a general view of the security posture of an organization. These include the number of registered targets, number of scans undertaken, risk distribution for all targets, and recently, number of assessment activities conducted. Graphical visualisations (charts and summary cards) can be used by

administrators to identify high-risk systems more quickly than to have to read the scan results for each system one at a time.

Other modules like target management, scan execution, report generation and scan history are also linked on the dashboard. The functions enable all the major security assessment tasks to be performed in a single interface. Centralized design eliminates the confusion of navigation between pages and makes it easier to use.

Overall, the Dashboard interface enhances the usability of the system by organizing and visualizing security information to make it clearer and easier to understand and facilitating efficient decision-making during security management processes.

4.3.2 Add Target Interface

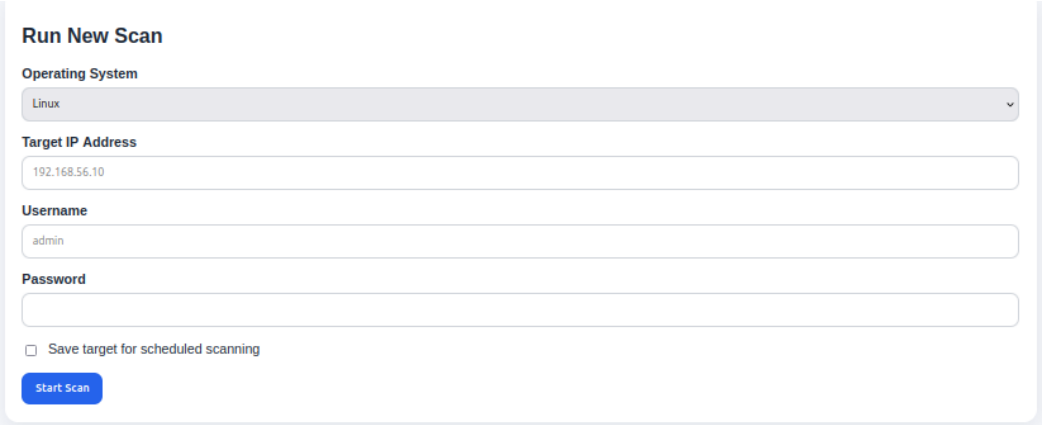


Figure 23: Add Target Interface

The Add Target Interface lets the administrators add new machines that will be included in future security assessments. Because the proposed system will be able to run on Linux and Windows OS, the interface has been designed to gather all the necessary information needed about the connection to the target device to achieve the safe remote communication.

If a new target machine is added, the administrator must input the IP address, the operating system type, the user name and authentication information. These information are checked prior to being saved to the database to decrease the chance of inappropriate settings or incomplete records. Input validation also helps to avoid connection failures in future scanning session.

This is especially significant if the operating system choice is going to affect which scanning module will be enabled. The Paramiko library is used to access Linux systems via SSH, and the pywinrm library is used to access Windows systems via WinRM (Windows Remote Management). The automatic module selection enables the administrator to control multiple operating systems from a single interface, without having to switch to various scanning methods.

Once successfully registered, a target machine is then available to be manually or scheduled for a security assessment. The centralized target management approach minimizes repetitive data entry and enables administrators to keep a tidy list of systems to be monitored for security.

The interface is therefore used as the base of proposed system as it ensures the proper number of all the target machines are registered in the system before any security assessment is carried out.

4.3.3 Scan Result Interface

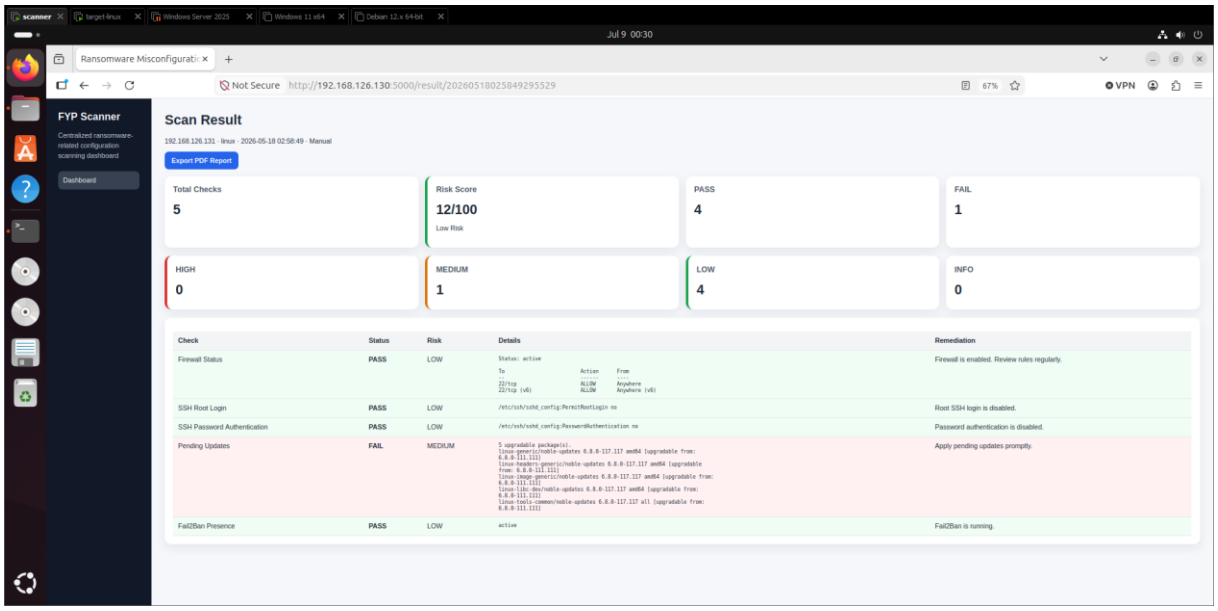


Figure 24: Scan Result Interface

The Scan Result Interface shows the results of each security assessment that has been completed. It gives the administrator a comprehensive overview of the security configuration status of a selected target machine and shows any possible weaknesses that could make it more vulnerable to ransomware attacks.

The individual security configuration check and the assessment status for each check is shown separately. To make it easier to read the system classifies each finding as PASS, FAIL, or INFO. The evaluated configuration meets the recommended security baseline (PASS) or the evaluated configuration does not meet the recommended security baseline (FAIL). INFO provides additional information that may help the administrator but would not directly impact the overall security assessment.

In addition to individual results, the site also shows the calculated risk score and risk classification overall. The numerical score allows administrators to understand the severity of the identified security issues, and the accompanying Low, Medium or High risk category gives a simplified overview for quicker decision making.

Remediation recommendations are shown along with failed security checks to further aid administrators. These recommendations are designed to explain what needs to be done to increase

the security configuration, like turning on the firewall, disabling unneeded services, improving the security for the password policy, or updating the operating system. This helps the administrator who is looking for security best practices without having to dig it up on their own.

The Scan Result Interface is thus not just a reporting page, but also a decision-support tool that enables administrators to prioritize corrective actions based on the severity of security weaknesses identified.

4.3.4 PDF Report Interface

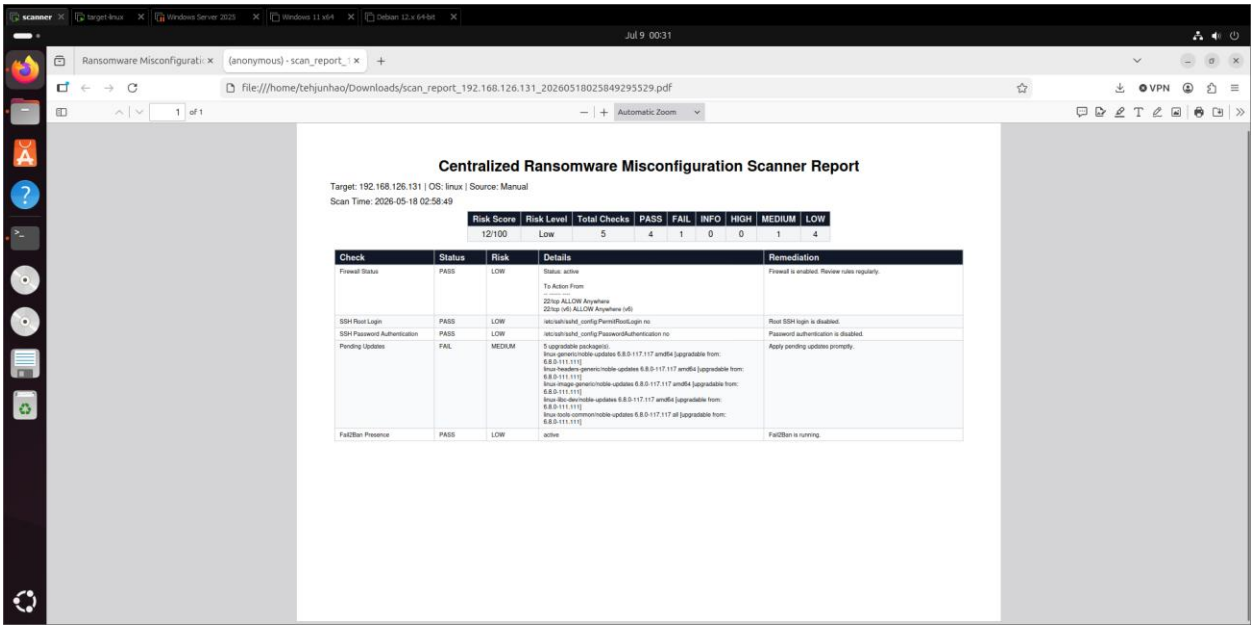


Figure 25: PDF Report interface

The PDF Report Interface provides the opportunity to send completed security assessments to a reportable and documented document format. Many organizations require security assessment records for various reasons, such as compliance reviews, internal audits, or historical reference. Therefore automating the generation of reports is more useful for the proposed system.

It extracts the related assessment details from the database, and formats it in a professional PDF. Details of the target machine, the scan date and time, detailed results of the security scan, the calculated risk score, overall risk level and recommended remediation are all included in the report generated. The results of an assessment will be consistent from a scan regardless of the person conducting the scan.

The PDF format also enables that assessment results can be handed over to the administration groups, auditors or any other IT personnel without needing to log on the scanning platform. The report provides all important security information in a portable format, so that organizations can store assessment results for comparison and security trend analysis.

In addition, having historical reports will help administrators track the progress after remediation. Organizations can assess their security position and see improvement when comparing reports from different assessment periods.

To sum up, PDF Report Interface enhances the proposed system to be more user friendly, as it converts the technical assessment data into a structured document that can be communicated, documented, and managed for future security.

4.4 Execution

4.4.1 Flask Dashboard Module

The Flask Dashboard Module will be the primary management console for the hoped for security configuration scanning system. The main reason for choosing Flask as the main web framework is that it is a lightweight, free and a good platform for building scalable web-based systems. Flask offers enough flexibility for multiple scanning modules integration with relatively simple application structure when compared to larger web frameworks (Welcome to Flask — Flask Documentation (3.1.X), 2026) Paramiko library to establish SSH connections.

The dashboard will be the communication link between all the back-end components and the administrator. The Flask application handles all the events of the users, including adding new target machines, running manual scans, scheduling automatic scans, checking the history of scans and creating PDF reports. The system allows for easier and more efficient use of admins, as these modules don't need to communicate directly with each other, but with the dashboard.

Requests from the admin are made using a web browser to the Flask server in a running system. Once the user's input is valid, the server passes the request on to the correct backend module. For example, if the admin initiates the scan on a Linux machine(s), then Flask will invoke the Linux scanning module. Likewise, Windows targets are automatically sent to Windows scanning module. After the scan is complete, Flask receives the assessment results back, stores these results in the database and updates the dashboard interface.

The dashboard also has graphical elements for data visualization, which enhances situational awareness. Administrators can quickly access information on vulnerable systems from the risk distribution chart and scan summary and historical assessment information without reading individual reports. This central management method has a great benefit in improving operation efficiency, especially in the SME environment, which has limited IT resources.

The Flask Dashboard Module is the backbone of the proposed system as it will be used to communicate all the software and it also provides a user-friendly interface to the administrators to manage the security assessment.

4.4.2 Linux Scanning Module

The Linux Scanning Module is used to assess the security settings of Linux target machines. The module initiates secure remote communications to the target system by means of the Secure Shell (SSH) protocol via the Paramiko Python library and gathers the information in a configuration without physically connecting to the target system (*Welcome to Paramiko! — Paramiko Documentation*, 2026).

The module will check if the target machine is reachable, and if it is, checks whether the authentication credentials provided are valid, before starting any security evaluation. In case of failure in authentication, the scanning is completed immediately, and an error message is returned to the administrator. This validation process will help avoid unnecessary executions and will allow for secure communication between the scanning server and the target machine.

Once successfully connected, the module runs some of the pre-defined checks that are directly linked with ransomware prevention. Some of these look at the status of the firewall, if the SSH root login is disabled, if password authentication is enabled, if there are security updates recently installed, or if Fail2Ban protection is available. The settings were chosen as they are often used in ransomware attacks if they are not set properly.

The results of each assessment are saved individually as PASS, FAIL, or INFO before being sent to the centralized risk scoring engine. The scanning module is not used to perform any calculations and only to collect data and perform basic evaluation, the final calculation of risk is done in the risk scoring module. Such segregation of responsibilities makes the system more modular and future versions of it could be secured with more Linux security products without impacting other parts of the system.

The Linux scanning module thus offers an effective and automated way to test the security of Linux configurations and to save the time that is normally needed for manual system evaluations.

4.4.3 Windows Scanning Module

The Windows Scanning Module is used to automatically conduct a security configuration assessment against a Microsoft Windows operating system. The communication between the centralized scanning server and target machine is made possible via the Windows Remote Management (WinRM) protocol with the use of pywinrm. This allows the administrators to carry out assessments remotely, without the need for in-person access to the targeted computer.

After connecting to the remote computer, the module checks a number of typical Windows security settings linked to ransomware attacks. These include the status of Windows Firewall, Remote Desktop Protocol (RDP), Microsoft Defender protection, password policy configuration, status of Administrator account, status of Guest account, and some other operating system security settings.

The collected configuration is then compared with security recommendations that are stored in predefined application logic. Any configuration that meets the recommended security baseline is rated as PASS and any that doesn't meet the security baseline is rated as FAIL. If the data is not used directly in scoring, but provides information that may help the risk score, it will be recorded as INFO. The standardized classification enables administrators to understand the results of the scans in a consistent way, across different operating systems.

The information gathered is sent to the centralized risk scoring engine which calculates the overall security position of the target machine. The modular design of the Windows scanning component also carries some advantages for future security checks, as more ransomware attack techniques are discovered.

The Windows Scanning Module allows administrators to scan Windows security configurations in an automated way from a remote location and ensure uniformity between several target systems.

4.4.4 Risk Scoring Engine

This is done within the Risk Scoring Engine to convert the individual security assessment results into a single score which reflects the overall security state of each targeted machine. The engine does not give administrators multiple separate findings, but instead provides an overview of the assessment that can be used to accelerate the decision-making process in a simpler risk score.

Once the Linux or Windows scanning module sends the scan results to the engine, it analyzes each security problem based on a set of prioritization rules. Disabled firewalls, weak password policies, exposed remote access services, or outdated operating systems are examples of critical vulnerabilities that could directly facilitate ransomware attacks and can add more risk points than less severe findings.

The total score is then assigned to one of three pre-established risk categories: Low Risk, Medium Risk, or High Risk. This classification offers an instant idea of the urgency of the security issues identified to the administrator. High Risk machines must be remediated immediately, and Low Risk systems conform to most of the recommended security guidelines(*CVSS V3.1 Specification Document*, 2026).

The engine not only calculates the overall score, but it also provides remediation suggestions for each of the failed configurations. these suggestions help administrators fix security gaps effectively without needing to possess in-depth cybersecurity expertise. The engine is not only a quantitative risk assessment but also provides practical information, making the proposed system more useful.

Risk Classification:

Score Range	Risk Level
0 – 39	Low
40 – 69	Medium
70 – 100	High

Figure 26: Risk Scoring Table

4.4.5 Scheduled Scanning Module

The Scheduled Scanning Module allows the proposed system to automatically execute security evaluations a predetermined time. The scheduling function takes care of regularly reviewing security configurations, which otherwise is only possible through manual action of the administrator, and thereby lowers the risk that new vulnerabilities are not identified for long periods of time(*OWASP Application Security Verification Standard (ASVS) | OWASP Foundation*, 2018).

Scanning schedules can be configured using the web dashboard by using the desired execution frequency with administrators. After it's been activated, the scheduling service will continuously check the set execution time and run assessments when the conditions are met. At execution time, each of the registered target machines is scanned with the same procedures as those used when conducting manual assessments, thus maintaining the consistency between scheduled and manual assessments.

Once the predetermined assessment has been accomplished, all scan results are automatically saved in the database. With the report generation turned on, the system also generates PDF reports on its own without administrator's intervention. These reports contain scan dates, security findings that are detected, calculated risk scores, and remediation suggestions which organizations can use to keep a history for compliance and auditing.

This automatic scheduling feature dramatically cuts routine security check administration. This is especially valuable for SMEs without dedicated cybersecurity staff, as the security monitoring can be done without constant manual intervention. The Scheduled Scanning Module therefore helps to ensure a more stable security posture over time and the overall efficiency of the proposed centralized scanning system.

4.5 Summary

This chapter introduced the design and implementation of the proposed centralized security configuration scanning system to detect ransomware related security configuration misconfigurations in SME environments. The system was designed with a modular architecture, combining various components to be managed in a centralized manner, based on the requirements identified during the investigation phase. Each module carries out a particular function and each of them collaborates with the others to complete a security assessment process efficiently and automatically.

The design section covered the overall architecture of the system, flow of the system, use case, a sequence diagram, a risk scoring mechanism, pseudocode for the Linux and Windows system scanning modules, the use of a scheduler, and the architecture of the database. These design artefacts demonstrate the interactions between the various parts of the system during scanning, from creating secure remote connections to producing security assessment reports. The modular design also facilitates better system maintainability and enables additional checks or even a new operating system to be added in the future's development without any big change in the existing architecture.

The chapter also explained about the interface designing of the proposed web application. The usability has been taken into consideration in all the following features to enable administrators to conduct security assessments using a central and easy to use platform: dashboard, target management page, scan result interface and PDF reporting function. The system offers a comprehensive view of security threats and provides administrators with recommendations for mitigating risks and strengthening security, including remediation suggestions.This allows administrators to gain a more complete understanding of their environment and prioritize corrective measures based on the severity of the threats found and the suggested remediation.

In addition, the major system modules were described, such as the web application (Flask) support modules (Linux and Windows) for scanning, risk scoring engine, and automated scheduling module. All these modules provide the ability to remotely assess the security configuration, analyze the detected misconfigurations, categorize the security risks, and produce detailed assessment reports. The incorporation of these elements highlights the potential of the

proposed solution to enable centralized security management while minimizing the need for manual security checks.

It could be concluded that the design and implementation of the program are complete and a sound basis for the effectiveness of the proposed system. The testing methodology, experimental results and discussion of the performance of the system in detecting ransomware related security misconfigurations for various scenarios will be discussed in next chapter.

Chapter 5

5.1 Introduction

The testing and evaluation of the proposed centralized security configuration scanning system is presented in this chapter. The main aim of this evaluation is to find out whether the developed system is working properly or not and whether it is meeting the project goals outlined in Chapter 1. The working, reliability and usability of the system was tested on a number of activities to ensure the system's functionality with respect to ransomware related security misconfigurations in both Linux and Windows operating systems.

In order to assess the effectiveness of the proposed solution two test methodologies were taken. The first is Black Box Testing, which is a test that checks to see if each piece of the system functions produces the appropriate output, but does not worry about the implementation of the source code. This testing method was chosen, because it tests the system as an administrator does and guarantees that all major functions work correctly in various situations.

The second approach is User Acceptance Testing (UAT) where the usability and feasibility of developed system have been tested. Some users engaged with the system, performing basic activities like registering target machines, running security scans, viewing security scans results, and creating PDF reports. Their input was gathered to see if the system performs as per the user's expectations regarding the functionality, ease of use, and overall effectiveness of the system.

In addition to testing individual functions of the system, this chapter examines the testing results and the discussion of how this proposed system accomplishes the project goals. The positive and negative aspects that were seen during the system evaluation are included together with the overall performance of the solution developed is presented. The results shown in this chapter offer an empirical basis for supporting the claims of the proposed centralized security configuration scanning system to effectively help administrators identify ransomware-related security misconfigurations in SMEs.

5.2 Testing Design

To validate the proposed system two complementary testing methodologies were used to validate the technical correctness and user satisfaction of the proposed system. The Black Box Testing was carried out to ensure that all the functional modules were performing as per the system requirements and the User Acceptance Testing (UAT) was performed to check the usability and feasibility of the developed application from the users point of view.

The testing activities were conducted in a simulated lab environment, which included a central Flask server and Linux and Windows target machines. The Linux operating system was remotely accessed via Secure Shell (SSH) library and the Windows operating system was remotely accessed via Windows Remote Management (WinRM) library. The test environment involved setting up a small enterprise network scenario that allowed multiple systems to be tested from a single management console.

To evaluate, a number of core system functions were chosen, all of which comprise the complete workflow of the proposed solution. They include registering the target machines, starting security scans, conducting Linux and Windows configuration assessment, calculating security risk scores, displaying scan results, creating PDF reports and running scheduled scans. There were pre-defined test cases used to evaluate each function to see if the output was what was expected vs the actual behaviour of the system.

User Acceptance Testing was conducted as well, with several participants using the finished system, to verify its functionality. The participants had to execute typical administrative tasks, and then perform an assessment of the system with regard to: Usability, Functionality, Interface Design, Report Quality and Overall Satisfaction. Technical testing along side user evaluation gives a complete evaluation of the proposed system effectiveness.

5.2.1 Testing Environment

Component	Specification
Web Framework	Flask
Programming Language	Python 3.x
Database	SQLite
Linux Connection	SSH (Paramiko)
Windows Connection	WinRM (pywinrm)
Operating Systems Tested	Ubuntu Linux, Windows 11
Report Format	PDF
Browser	FireFox

Figure 27: Testing Environment Table

5.2.2 Testing Methods

Two tests were chosen to thoroughly test the proposed system.

Black Box Testing

Black Box Testing is concerned with the correctness of the system developed with regard to its functional requirements. This testing method tests the system's behavior by giving input to the system and seeing the output that is produced, without looking at the way in which the program is implemented in practice. As administrators use the application via the GU interface rather than the source code, Black Box Testing is an effective representation of real-world usage situations(*The Art of Software Testing*, 2026).

In the evaluation, all the key components of the system were tested such as target registration, Linux scanning, Windows scanning, risk score calculation, report generation, and scheduled scanning. The successful completion of each function is an evidence of the fulfillment of the functional requirements of the application developed.

User Acceptance Testing (UAT)

To find out if the finished system is ready for the intended users to use, user acceptance testing was carried out. The participants were invited to use the app on their own, and were tasked with doing a few basic security management activities. Following the evaluation, participants gave their feedback on the usability, functionality, interface design, response time, and usefulness of the reports produced(Pressman, 2014).

All of the collected feedback was then analysed to see if the security configuration scanning centralisation system that was suggested would be acceptable to users of a similar environment to that of an SME. The results also help identify potential enhancements in the future development of the system.

5.3 Black Box Testing Result

Black Box Testing has been performed to check whether each functional module of the proposed system is working as per the requirement of the system or not. In this evaluation, the internal program code was not taken into account. Rather, a testing was conducted to ensure that appropriate user inputs were given and the output from the system was observed. The testing approach is based on the reality of administrators functioning with the system via the graphical user interface.

All the major features of the proposed system were tested such as target registration, security scan on Linux system, security scan on Windows system, calculation of risk score, generation of PDF report, and scheduled security scan. The functions were performed multiple times with varying test cases to ensure the consistency and reliability of the system.

Test ID	**Function**	**Test Scenario**	**Expected Result**	**Actual Result**	**Status**
BB01	Add Target	Register a Linux target with valid credentials	Target machine successfully added	Target added successfully	Pass
BB02	Add Target	Register a Windows target with valid credentials	Target machine successfully added	Target added successfully	Pass
BB03	Linux Scan	Execute security scan on Linux machine	Scan completes and displays findings	Scan completed successfully	Pass
BB04	Windows Scan	Execute security scan on Windows machine	Scan completes and displays findings	Scan completed successfully	Pass
BB05	Risk Scoring	Calculate overall risk level	Risk score calculated correctly	Correct risk score displayed	Pass

BB06	PDF Report	Generate assessment report	PDF report generated successfully	Report downloaded successfully	Pass
BB07	Scheduled Scan	Execute automatic scheduled scan	Scan starts automatically at scheduled time	Automatic scan executed successfully	Pass

Figure 28: Black Box Testing Table

The test results shown in Table 5.1 have shown that all the major system functions have been successful during the tests. All test cases gave the desired output with no functional failures. This proves that the application developed meets the main functional requirements and can be used to carry out security configuration evaluations in a centralized way on various operating systems.

The Add Target module was able to successfully validate administrator input before storing target information to the database. Future manual and scheduled assessments could be carried out efficiently as there was no error during the registration of both the Linux and Windows machines.

The Linux and Windows scanning modules were also stable during the tests. Both modules were able to successfully connect remotely, gather security configuration data, analyze security settings, and report back to the Flask dashboard. When proper authentication credentials were provided, no communication failures were seen.

Additionally, the Risk Scoring Engine accurately determined a risk score from the security findings it had identified. Machines that had more than one vulnerability were consistently identified with a greater risk score, and systems that were configured securely had a proper Low Risk score. This verifies that the scoring algorithm is working as per the evaluation rules as defined in the system implementation.

The PDF reporting feature also worked as it should, producing reports on security assessments that included scan results, time, risk categories, and remediation tips. These reports give administrators a permanent record to show compliance and for future security reviews.

In general, the Black Box Testing results show the proposed system is reliable and successful in incorporating all major modules together as a security management system.

5.4 User Acceptance Testing (UAT)

Black Box Testing is used to verify that a system is correct in terms of functions, while User Acceptance Testing (UAT) checks that the final system meets the requirements and expectations of its end-users. Given that the proposed system is intended for use by SME administrators and cybersecurity practitioners, it is equally crucial as the technical function that the system be user friendly.

Five participants (with backgrounds in cybersecurity, information technology and system administration) were invited to review the completed system. Each participant had to execute a number of common events such as register a target machine, start a security scan for Linux and Windows OS, analyze results of the security scans, interpret the risk score calculated by the security scan, and create a pdf report.

Participants answered the following questions after performing the activities, in order to assess the system, using a 5-point Likert scale from Strongly Disagree (1) to Strongly Agree (5).

Evaluation Criteria	**Average Score**
The system is easy to use	4.8
The dashboard is well organized	4.6
Security scan results are easy to understand	4.8
Risk score is useful for decision making	4.8
PDF report is informative	4.6
System performance is satisfactory	4.4
Overall satisfaction	**4.7**

Figure 29: User Acceptance Testing Result

The User Acceptance Test results show that the proposed system has high level of user satisfaction. The average score of the participants overall was 4.7 out of 5, indicating that the application was found to be practical and appropriate for practical security management tasks.

The dashboard interface and centralized management system were especially well received by the participants. Users could carry out security evaluations from one web interface as opposed

to having to connect to multiple machines. This cut down the amount of time needed to review several systems and enhanced the user experience.

The risk scoring feature was among the other highly recommended features. The participants said that a combined presentation of findings and overall risk level was helpful to prioritizing remediation activities. Administrators were not required to read through a lot of technical details; they could see at a glance which systems needed immediate attention.

Positive feedback was also received for the automatically generated PDF reports that were able to provide a structured summary of the results of the assessment which were suitable for documentation and auditing. The participants noted that the reports could be made available to management teams or stored as evidence in compliance reviews in the future.

The overall evaluation was quite positive, but a number of suggestions were offered as to possible improvements. They included adding more operating systems, expanding role-based access control, adding email notifications on scheduled scans, and more. Here are some recommendations to guide future system improvements.

The User Acceptance Testing (UAT) generally validates the functionality and usability of the proposed centralized system for scanning security configurations, which can be deployed in SME environments.

5.5 Discussion of Results

The Black Box Testing and User Acceptance Testing have been conducted, and the result shows that the proposed centralized security configuration scanning system is able to perform the expected functions and meets the objectives set at the start of this project. All major system modules were functional during the test and no critical functional errors occurred. The execution of the testing activities was successful, which means that the system is stable and can be used to support security configuration assessments in SME environments.

The Black Box Testing ensured that all the basic functionalities of the application gave the expected output. Target registration was successful, as was the scanning of Linux and Windows targets, the calculation of the risk scores, the generation of PDF reports and the scheduled scanning, which were all done on the evaluation. These modules were integrated successfully, which indicates the reliability of the overall system architecture and the proper functioning of the communication between the different software components. The product is a collection of separate modules integrated into a single application, so it is critical that each of them is able to communicate with each other.

The Linux and Windows scanning modules both had the ability to connect securely remotely with SSH and WinRM respectively. After connecting the modules, they were able to collect configuration data and analyse the preset security settings, and to send the results to the Flask dashboard. Supporting both operating systems offers an additional degree of flexibility for SMEs, as many businesses have a heterogeneous mix of Linux and Windows devices. This cross-platform feature makes the proposed system more useful than systems that operate on a single platform.

The risk scoring system was also stable during the testing process. The higher the risk score, the less secure the configuration was, and the lower the risk score the more secure the configuration was, with those that were configured according to the recommended security practices being designated as Low Risk. This proves that the pre-defined scoring algorithm is able to differentiate between secure and vulnerable systems. The overall risk score will give an administrator a quick and easy indicator of which machines need their attention, rather than having to manually analyse

every individual finding. This feature makes decisions easier, and allows administrators to prioritize remediation activities more effectively.

The system is also found to be practical through User Acceptance Testing. The overall satisfaction measures found that the participants think the application is easy to learn, simple to use, and helpful for conducting regular security assessment tasks. The centralized dashboard was a valuable feature participants found to be useful; this eliminated the necessity to manage multiple devices. The graphical display of scan results and the addition of remediation suggestions also made assessment reports easier to read, enabling users with varying degrees of cybersecurity familiarity to better digest and comprehend the results.

The test results were mostly favourable but some shortcomings were noted in the evaluation of systems. This current implementation is mainly based on Linux and Windows OS and is not applicable to other operating systems like macOS or network devices. Furthermore, the risk scoring system is also rule-based (not adaptive) at present. Future risk assessments may be even more accurate if they were also powered by machine learning or threat intelligence, as cybersecurity threats continue to use new methods.

In general, these test results show the proposed system's ability to successfully detect security misconfigurations that are associated with ransomware, which can be managed centrally. By integrating automated scanning, risk assessment, report generation and scheduled monitoring, administrators can optimize the efficiency of security management while also minimizing the amount of manual work that goes into regular security inspections.

5.5.1 Achievement of Project Objectives

To see how well the proposed system met the project objectives set out in Chapter 1, the performance of the proposed system was compared to the project objectives.

Objective 1

- To detect ransomware configuration security issues in Linux and Windows systems.

This goal has been accomplished. The implemented system was able to remotely connect to Linux and Windows target machines by secure communication protocols and automatically conduct security configuration evaluations. Testing was successful on multiple security settings such as firewall configuration, password policies, remote access settings, operating system updates, and security services. The system always detected insecure configurations and reported them on the centralized dashboard.

Objective 2

- To create a centralized web based platform for configuration management in security.

This goal was accomplished as well. The Flask based dashboard combined all the major functions of the system into a single web interface. Using a single interface, administrators could register target machines, run manual or scheduled scans, view assessment history, compute risk scores, and create PDF reports. Its centralized system made it easier to manage the system and less complex to perform security assessments on several devices.

Objective 3

- To perform risk assessment and risk remediation suggestions to enhance system security.

Risk scoring engine and remediation recommendation module have been implemented to achieve the third objective. The system was able to determine an overall security risk score for each assessment, and assign a target machine to be Low, Medium or High Risk after the assessment. The security weaknesses identified were helpful to administrators in recognizing this and then gave suggestions on how to make better configurations. These capabilities allow organizations to prioritize corrective actions more effectively and minimize ransomware attacks.

Based on the overall testing results, all three project objectives have been successfully achieved. It is shown that the proposed system can be used to assist SMEs to enhance their cybersecurity postures, with less effort spent on routine security monitoring.

5.6 Summary

The testing and evaluation of the proposed centralized security configuration scanning system have been outlined in this chapter. Through the process of Black Box Testing it was found that all major functions of the system worked as per the stated functional requirement and User Acceptance Testing was conducted to test the usability and usability of the application from the user's point of view. The test results showed the developed system could reliably assess the security configuration at a central point on both Linux and Windows operating systems. The result of the test performed indicated that the proposed system could integrate the remote scan, automated risk assessment, centralized management, PDF report generation and scheduled scanning into one system. Other user feedback shared also included that it is easy to use, it offers valuable security information, and helps administrators spot ransomware security misconfigurations efficiently. While a number of improvements were suggested, the overall assessment indicates that the system developed meets the project requirements and fulfills all project objectives. The results obtained in this chapter confirm that the proposed solution is technically working, practically useful and applicable in supporting the security configuration management task in SME environment. This chapter wraps-up the project with an overall summary of the project, notes on the challenges faced while developing the project and recommendations for further improvements.

Chapter 6

6.1 Conclusion

The project aimed to create a centralized security configuration scanning system that identifies security misconfigurations that are indicative of ransomware attacks in Small and Medium Enterprise (SME) environments. This system was developed due to the rising number of ransomware attacks against organizations due to weak system configurations, not software vulnerabilities. Most SMEs don't have a specific budget for cybersecurity, and they are also frequently stuck with manual security checks, which are labor-intensive, error prone and inconsistent. Hence, the proposed system aimed to give an automated system and centralized solution for the security posture of organizations, where the security posture could be assessed in a simplified way.

The project started with extensive research to gain insight into the state of the ransomware threat landscape, the state of existing security configuration management practices, and limitations of the security configuration assessment tools that are available. Literature review and survey data helped to obtain the requirements of the users and design the proposed system. From this information, a web-based centralized platform was created with the Flask framework that includes Linux and Windows security configuration scanning and combines them into one management interface.

The completed system is able to send and receive security configuration assessments remotely using SSH and WinRM communication protocols. The system gathers security configuration data on target machines and analyzes each finding, determines a total security risk score, categorizes the system into defined security risk tiers, and creates remediation suggestions for system administrators to help enhance the security of their systems. The capability of creating PDF reports and allowed scheduled scanning further adds to the usability of the created solution in day-to-day security management tasks.

System evaluation showed that the system proposed is reliable in the selected testing environment. The results of Black Box Testing were that all major system functions performed as intended and the results from User Acceptance Testing showed that the participants found the application easy to use, practical and suitable for conducting routine security assessments. The

system's centralized management, automated scanning, risk scoring, and reporting indicate that it can help administrators understand their security vulnerabilities better than with traditional manual techniques.

Overall, the project has been successful in accomplishing its goal and objectives. The system developed in this paper shows that a centralized security configuration assessment can enhance the efficiency of security management and decrease the operational effort to be spent in detecting common ransomware configuration-related risk in SME environments.

6.2 Project Contributions

The proposed centralized security configuration scanning system offers a number of practical benefits to both SMEs and the cyber security community. The developed application, from an organizational standpoint, is a lightweight and centralized application, which allows the administrator to manage multiple systems in a single management platform. This helps to minimize the need for manual inspection tasks to be repeated and enhance operational efficiency, especially for companies that don't have a large cybersecurity team.

Automated risk scoring and remediation recommendations enable administrators to prioritize remediation tasks according to the severity of the detected security issues. The calculated risk level allows administrators to more rapidly determine which systems need immediate attention rather than examine multiple technical findings individually. This increases the speed of decisions and more efficient deployment of resources in security management operations.

The project also illustrates how several open source systems can be integrated into a single cyber security management system and how these systems can be integrated with each other, such as Flask and Paramiko, WinRM and PDF reporting. The implementation can be used as a guideline by future researchers and students in the field of building security assessment tools with open source software.

Academically, the project advances the knowledge of ransomware prevention by concentrating on security configuration management and not just on detecting ransomware. The results back up the notion that system hardening can have a significant impact on the attack surface that ransomware actors can exploit. As a result, the project is promoting a more proactive attitude to cyber security management of organizations.

6.3 Limitations

The system proposed was able to meet the main goals, but during the development and evaluation process some limitations were found.

The first is that the current implementation only works for Linux and Windows operating systems. The current release does not support other widely-used platforms like macOS, network appliances, and cloud-based infrastructure. Organizations running heterogeneous environments might then need extra evaluation tools to enable a total coverage of the infrastructure.

Secondly, security configuration assessments are based on pre-designed security rules that were set during system implementation. These rules are effective against detecting common ransomware-related misconfigurations, but may not be effective against new attack techniques or new security vulnerabilities. Security threats are continually changing, and the rule database will need to be updated from time to time.

Another drawback is the risk scoring system. The current implementation utilizes a rule-based scoring model and assigns a weight to each individual security finding. This method has limitations as it does not take into account organizational priorities, business impact, or real-time threat intelligence. Therefore, the scoring models may need to be tailored to reflect the operation of different organizations.

In addition, these testing activities were done in a controlled virtual laboratory instead of in a production network. This method allowed for a consistent means of evaluation, but it could present other operational issues if it is expanded into larger enterprise networks with hundreds of devices, network segmentation, or more stringent access control policies.

Last, only a few people were involved in UAT. The feedback received was favourable, but a larger and more varied set of users would result in greater conclusive evidence of the system's usability and effectiveness in various organisational contexts.

6.4 Recommendations for Future Work

To improve the capabilities of the proposed system in future development the following improvements can be considered:

One way to improve it is to add more operating system support such as macOS, cloud VMs, containers and network infrastructure devices. Enabling platform support will make the system more broadly applicable in the current enterprise landscape with a variety of operating systems.

Additional versions may also be able to include machine learning techniques to enhance the accuracy of security risk assessment. In addition to pre-defined scoring criteria, intelligent models may examine past scan results and recognize unusual patterns in configuration and automatically apply a priority to findings based on new ransomware attack types.

Another useful addition is incorporating external threat intelligence feeds like CVE databases, MITRE ATT&CK knowledge base or vulnerability intelligence platforms. The system could be used to incorporate and incorporate present threat data with configuration assessment to give greater accuracy in threat prioritization and propose remediation activities based on active ransomware campaigns.

Another enhancement to the notification mechanism could be the automatic sending of an e-mail or mobile message when a scheduled scan detects High Risk systems. This would allow administrators to act quickly when needed when there is a security crisis, without having to sit on the dashboard all day.

Last but not least, role-based access control, multi-user authentication, audit logging and cloud deployment are all possible future features. The enterprise-level features would enhance the system's scalability, security management capabilities and make it more appropriate for large-scale enterprises.

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APPENDICES

Appendices A: FYP Poster - A3 size (color poster)



Figure 30: FYP poster

Appendices B: PPF - Design and Evaluation of a Centralized Ransomware-Oriented Security Configuration Scanner for SME Networks Status

Project Title: Design and Evaluation of a Centralized Ransomware-Oriented Security Configuration Scanner for SME Networks

Status

APPROVED

Proposed By

TEH JUN HAO

Description

Introduction

Modern organizations rely heavily on information systems to support daily business operations, data storage, and communication. As these systems grow in scale and complexity, maintaining secure and consistent system configurations becomes an essential aspect of cybersecurity management. Security configuration refers to the process of defining, applying, and maintaining system settings—such as authentication policies, access controls, network services, and logging mechanisms—in a manner that reduces exposure to security threats while supporting operational requirements.

In practice, security configuration management is often performed through a combination of policy guidelines, administrative procedures, and technical controls. Larger organizations typically adopt centralized tools or frameworks to standardize configuration assessment and enforcement across multiple systems. Centralized configuration assessment enables administrators to remotely collect configuration evidence, compare system states against defined security baselines, and identify deviations that may introduce risk. This approach improves visibility, consistency, and accountability compared to manual and system-by-system checks.

Small and medium-sized enterprises (SMEs), however, often face challenges in implementing such centralized security practices. Due to limited financial resources, manpower, and specialized expertise, SMEs commonly rely on manual, ad-hoc, or irregular security checks conducted by IT administrators who manage multiple responsibilities simultaneously. As a result, configuration drift and unnoticed weaknesses may accumulate over time, increasing the organization's exposure to security incidents. While enterprise-grade vulnerability and configuration management platforms exist, they are frequently complex, costly, and misaligned with the operational constraints of SMEs.

Within the broader cybersecurity landscape, ransomware has emerged as one of the most disruptive threats affecting organizations of all sizes. Although modern ransomware variants employ advanced encryption and evasion techniques, successful attacks often depend on weaknesses in system and network configurations rather than purely on malware sophistication. Attackers typically exploit insecure remote access services, weak authentication mechanisms, excessive administrative privileges, insecure file sharing configurations, insufficient patching, and limited system logging to gain initial access, escalate privileges, move laterally within networks, and ultimately deploy ransomware at scale.

These observations highlight the importance of proactive and systematic security configuration assessment as a defensive measure against ransomware threats. Rather than reacting to incidents after compromise, organizations—particularly SMEs—benefit from tools that can continuously and consistently identify risky configurations before they are exploited. In response to this need, this project proposes the design and evaluation of a centralized security configuration scanning tool focused specifically on identifying misconfigurations commonly associated with ransomware attack paths. The proposed system aims to provide a practical, lightweight, and evidence-based approach to improving ransomware defensive readiness within SME network environments.

Aim

To design, implement, and evaluate a centralized security configuration scanning tool that detects ransomware-relevant misconfigurations in a networked environment, provides actionable remediation guidance, and demonstrates measurable risk reduction through before-and-after validation experiments.

Objectives

- To identify and justify a practical set of security configuration checks that are commonly associated with ransomware attack vectors, including remote access exposure, authentication and password policies, privilege management, file sharing configurations, firewall status, and system logging readiness.
- To implement centralized remote scanning from a dedicated scanning machine to multiple target systems using WinRM and PowerShell Remoting for Windows systems and/or SSH for Linux systems, with a focus on read-only evidence collection.
- To generate both machine-readable and human-readable reports, such as JSON outputs and HTML or PDF-ready summaries, containing identified issues, supporting evidence, and actionable remediation recommendations.
- To construct a virtualized laboratory environment and intentionally introduce representative security misconfigurations to enable controlled testing and validation of the scanning tool.
- To evaluate the effectiveness of remediation actions by comparing pre-remediation and post-remediation scan results, including reduction in critical findings, improvement in overall risk score, and changes in the feasibility of a simulated ransomware attack path.
- To document operational requirements, privilege assumptions, system limitations, and ethical and safe-use considerations, ensuring that all testing is conducted in an isolated environment without the use of real organizational identifiers or sensitive data.

Targeted Users

The proposed tool is primarily targeted at the following user groups:

1) SME IT Administrators and System Engineers who require a lightweight and cost-effective solution to identify, prioritize, and remediate risky security misconfigurations across servers and endpoints without deploying complex enterprise platforms.

Figure 31: PPF page1

posture, track remediation progress, and support internal security reviews.
4) Students and Cybersecurity Learners who seek a practical and hands-on project that demonstrates core security engineering concepts, including secure configuration assessment, remote system administration, risk prioritization, and measurable security evaluation.

SDG

SDG9

Keywords

Information security	Network Forensics
Forensic Analysis Tools	Cybersecurity
Network Vulnerability Analysis	Network Vulnerability Analysis
Cybersecurity	Cybersecurity

Preferred Supervisor(s)

SHAHAB ALIZADEH, NORIS BINTI ISMAIL, DR. HASSAN JAMIL SYED, SIVAKAMENI A/P INDRAN,

Assigned Supervisor

hassan.jamil

Assigned Second Marker

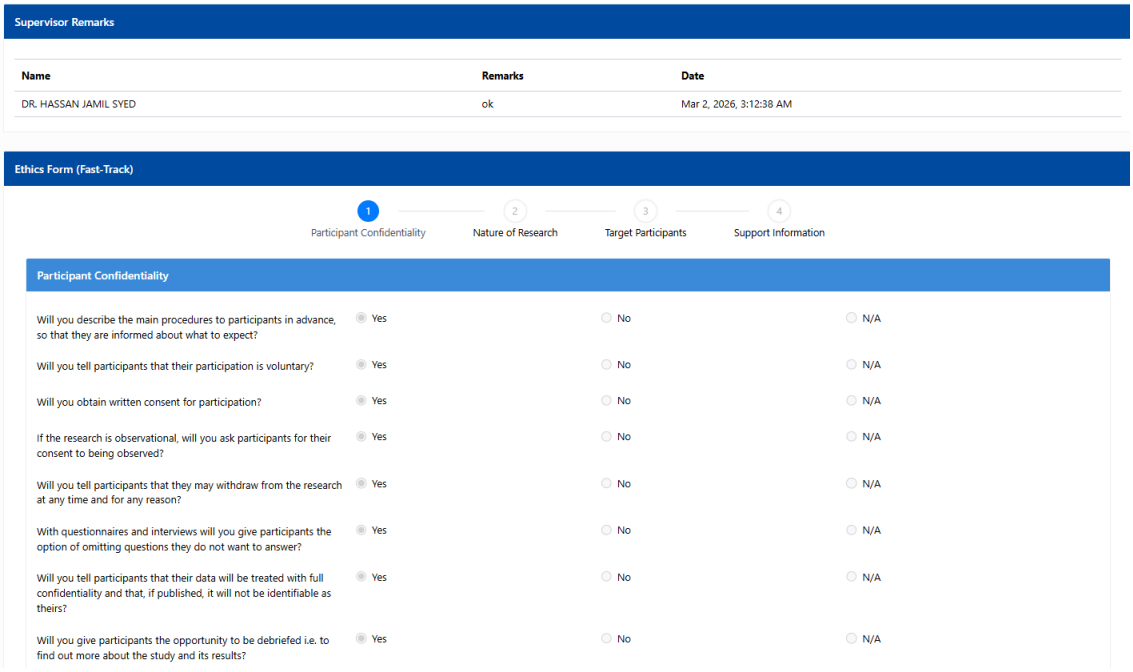
yogeswaran.nathan

Remarks

Name	Remarks	Date
TS. UMAPATHY EAGANATHAN	Good.	Jan 4, 2026, 12:32:28 PM

Figure 32: PPF page 2

Appendices C: Ethics Forms



Supervisor Remarks

Name	Remarks	Date
DR. HASSAN JAMIL SYED	ok	Mar 2, 2026, 3:12:38 AM

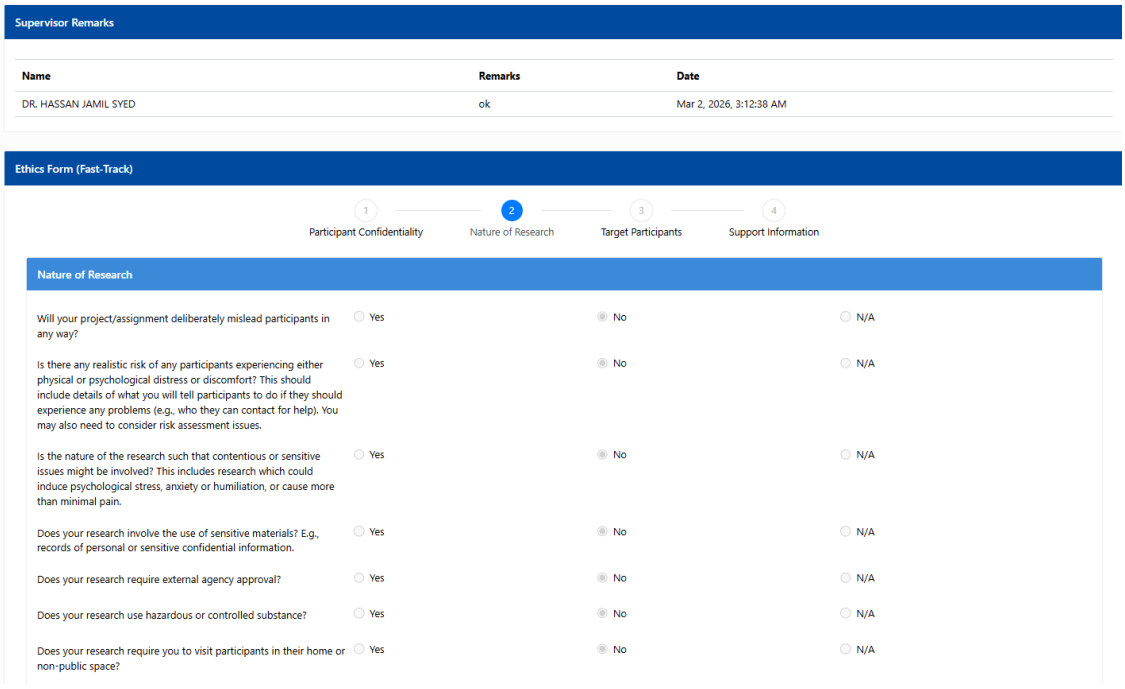
Ethics Form (Fast-Track)

1 Participant Confidentiality — 2 Nature of Research — 3 Target Participants — 4 Support Information

Participant Confidentiality

Will you describe the main procedures to participants in advance, so that they are informed about what to expect?	Yes	No	N/A
Will you tell participants that their participation is voluntary?	Yes	No	N/A
Will you obtain written consent for participation?	Yes	No	N/A
If the research is observational, will you ask participants for their consent to being observed?	Yes	No	N/A
Will you tell participants that they may withdraw from the research at any time and for any reason?	Yes	No	N/A
With questionnaires and interviews will you give participants the option of omitting questions they do not want to answer?	Yes	No	N/A
Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?	Yes	No	N/A
Will you give participants the opportunity to be debriefed i.e. to find out more about the study and its results?	Yes	No	N/A

Figure 33: Ethics Form page 1



Supervisor Remarks

Name	Remarks	Date
DR. HASSAN JAMIL SYED	ok	Mar 2, 2026, 3:12:38 AM

Ethics Form (Fast-Track)

1 Participant Confidentiality — 2 Nature of Research — 3 Target Participants — 4 Support Information

Nature of Research

Will your project/assignment deliberately mislead participants in any way?	Yes	No	N/A
Is there any realistic risk of any participants experiencing either physical or psychological distress or discomfort? This should include details of what you will tell participants to do if they should experience any problems (e.g., who they can contact for help). You may also need to consider risk assessment issues.	Yes	No	N/A
Is the nature of the research such that contentious or sensitive issues might be involved? This includes research which could induce psychological stress, anxiety or humiliation, or cause more than minimal pain.	Yes	No	N/A
Does your research involve the use of sensitive materials? E.g., records of personal or sensitive confidential information.	Yes	No	N/A
Does your research require external agency approval?	Yes	No	N/A
Does your research use hazardous or controlled substance?	Yes	No	N/A
Does your research require you to visit participants in their home or non-public space?	Yes	No	N/A

Figure 34: Ethics Form page 2

Does your research investigate illegal activities or behaviours?	Yes	No	N/A
Does your research involve discussion or collection of information on potentially sensitive, embarrassing or distressing topics, administrative or secure data? This includes research involving respondents through internet where visual images are used, and where sensitive issues are discussed	Yes	No	N/A
Will your participants be receiving financial compensation for participating in your research?	Yes	No	N/A
Will your research data be used in the future after the conclusion of your project?	Yes	No	N/A
Will your research involve in processing sensitive data belonging to an organisation/persons?	Yes	No	N/A
Will your research be collecting photographs, videos, and audio recordings of the participants?	Yes	No	N/A
Will the participants' personal particulars be known to any third party?	Yes	No	N/A
Will the participants' data confidentiality be made known to the public?	Yes	No	N/A
Will the research be conducted where the safety of the researchers maybe in question?	Yes	No	N/A
Will be the research be conducted outside of Malaysia and/or UK?	Yes	No	N/A

Figure 35: Ethics Form page 3

Supervisor Remarks

Name	Remarks	Date
DR. HASSAN JAMIL SYED	ok	Mar 2, 2026, 3:12:38 AM

Ethics Form (Fast-Track)

1 Participant Confidentiality — 2 Nature of Research — 3 Target Participants — 4 Support Information

Target Participants

Do participants fall into any of the following special groups? • Children (under 18 years of age) • People with communication or learning difficulties • Patients • People in custody • People who could be regarded as vulnerable • People engaged in illegal activities (e.g., drug taking) • Groups of people whose relationship among each other allow one to have influence over the other such as: Carers and patients with chronic conditions; teachers and their students; prison authorities and prisoners; employers and employees • Groups where permission of a gatekeeper is normally required for initial access to members.	Yes	No	N/A
Does the project/assignment involve external funding or external collaboration where the funding body or external collaborative partner requires the University to provide evidence that the project/assignment had been subject to ethical scrutiny?	Yes	No	N/A

Note: You may also need to obtain satisfactory clearance from the relevant authorities.

Figure 36: Ethics Form page 4

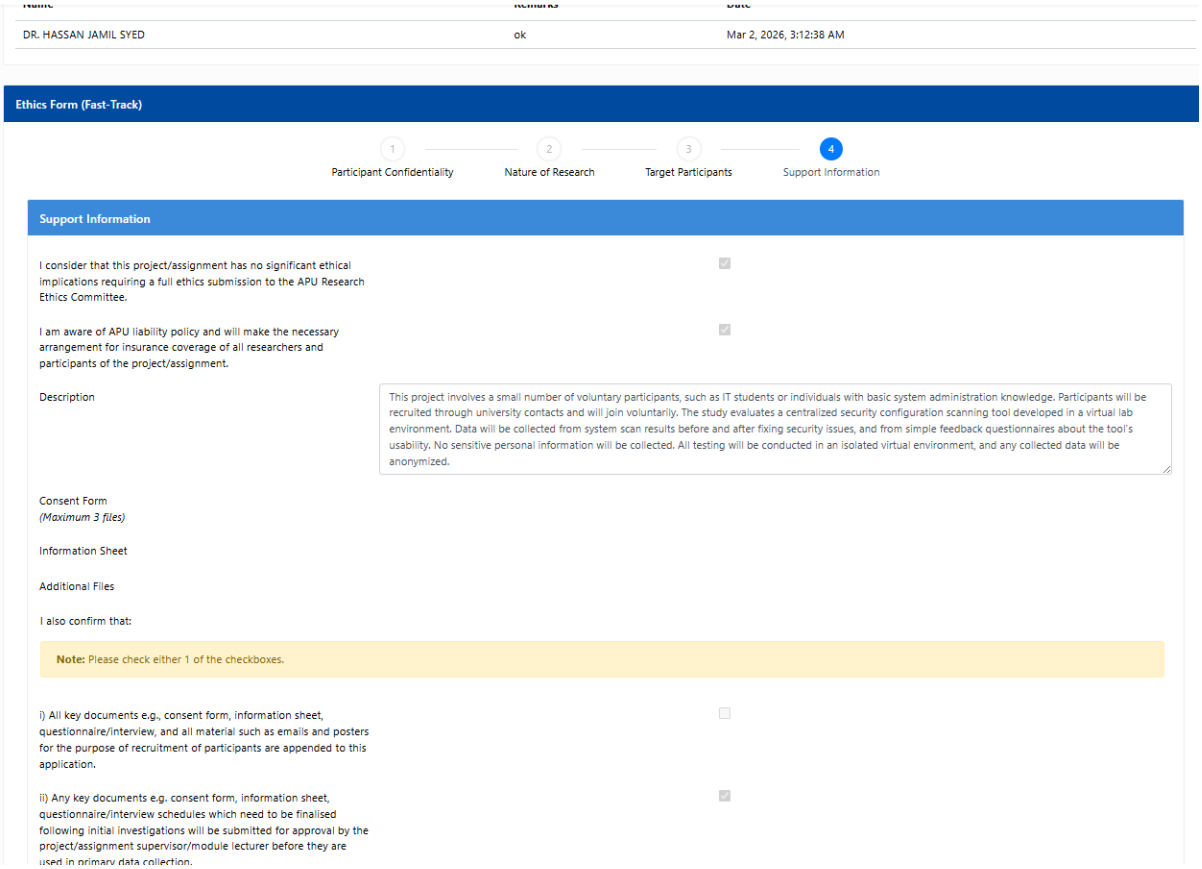


Figure 37: Ethics Form page 5

Appendices D: Log Sheets

Meeting Log Sheet

Student's Name: Teh Jun Hao Date: 05 Feb 2026 13:00 Meeting No: 1

Project tittle:

Centralized security configuration scanning tool to detect ransomware-related misconfigurations in SME environments

Intake: APU3F2511CS(CYB)

Supervisor's Name: Dr. Hassan Jamil Syed **Supervisor's signature**

Item for discussion (Noted by student before mandatory supervisory meeting):

1. Present the proposal of FYP project with supervisor.
2. Explains the Objectives, aims and goals to supervisor.
3. Get the approval from supervisor for the project proposal.

Record of discussion (noted by student during mandatory supervisory meeting):

1. Supervisor recommended to do more research related to my project topic
2. Supervisor recommended some useful tools for research purpose
3. Supervisor approved my FYP proposal and provided several recommendation
4. Supervisor provided guidance on the next steps and explained what should I do and prepared for the next meeting.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. Do more research for better understanding related of my project to current societal issues.
2. Start doing Chapter2 (Literature Review) based on the research findings.

Figure 38: Meeting Log Shhet page 1

Meeting Log Sheet

Student's Name: Teh Jun Hao Date: 27 Feb 2026 12:15 Meeting No: 2

Project tittle:

Centralized security configuration scanning tool to detect ransomware-related misconfigurations in SME environments

Intake: APU3F2511CS(CYB)

Supervisor's Name: Dr. Hassan Jamil Syed **Supervisor's signature**

Item for discussion (Noted by student before mandatory supervisory meeting):

1. Get the approval from supervisor for the survey questions
2. Get the meeting log sheet from supervisor
3. Present my research finding

Record of discussion (noted by student during mandatory supervisory meeting):

1. Supervisor approved my survey question
2. Supervisor reminded that all the finding, all the sentence must base on my own understanding

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. Proceed IR (Chapter 3) after collect 30 responders for the survey
2. Complete IR for final checking in the next meeting

Figure 39: Meeting Log Sheets page 2

Meeting Log Sheet

Student's Name: Teh Jun Hao Date: 09 Mar 2026 15:00 Meeting No: 3

Project tittle:

Centralized security configuration scanning tool to detect ransomware-related misconfigurations in SME environments

Intake: APU3F2511CS(CYB)

Supervisor's Name: Dr. Hassan Jamil Syed **Supervisor's signature**

Item for discussion (Noted by student before mandatory supervisory meeting):

1. Show completed IR to supervisor for final checking
2. Confirm the procedures for the submission of IR

Record of discussion (noted by student during mandatory supervisory meeting):

1. Provided clear steps of submission of IR
2. Supervisor reminded that all the finding, all the sentence must base on my own understanding

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. Fully complete the IR
2. Prepare all appendix required document or screenshot
3. Get the signed meeting logs from supervisor

Figure 40: Meeting Log Sheets page 3

Meeting Log Sheet

Student's Name: TEH JUN HAO **Date:14 July2026 12 pm Meeting No:** 4

Project tittle:

CENTRALIZED SECURITY CONFIGURATION SCANNING TOOL TO DETECT RANSOMWARE-RELATED MISCONFIGURATIONS IN SME ENVIRONMENTS

Intake:

Supervisor's Name: Dr. Hassan Jamil Syed **Supervisor's signature**

Item for discussion (Noted by student before mandatory supervisory meeting):

1.Final Year Project (FYP) progress overview

2.Documentation structure and content

3.Submission requirements and timeline

Record of discussion (noted by student during mandatory supervisory meeting):

1.Student presented overall FYP progress and explained the project objectives and scope.

2.Supervisor reviewed the documentation structure and provided feedback on clarity and completeness.

3.Submission requirements, report readiness, and final deliverables were discussed.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1.Refine and finalize project documentation based on feedback.

2.Prepare final submission materials for the FYP (POSTER, VIDEO RECORDING, DEMONSTRATE SYSTEM).

Figure 41: Meeting Log Sheets page 4

Meeting Log Sheet

Student's Name: TEH JUN HAO **Date:** 16 July 2026 12 pm **Meeting No:** 5

Project tittle:

CENTRALIZED SECURITY CONFIGURATION SCANNING TOOL TO DETECT RANSOMWARE-RELATED MISCONFIGURATIONS IN SME ENVIRONMENTS

Intake:

Supervisor's Name: Dr. Hassan Jamil Syed **Supervisor's signature**

Item for discussion (Noted by student before mandatory supervisory meeting):

1.Final Year Project (FYP) progress overview

2.Present Poster

3.Demonstrate System

Record of discussion (noted by student during mandatory supervisory meeting):

1.Student presented overall FYP progress.

2.Supervisor reviewed the system and provided feedback on clarity and completeness.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1.Refine and finalize project documentation based on feedback.

2.Prepare final demonstration of the system due some technical issue happened.

Figure 42: Meeting Log Shhets page 5

Meeting Log Sheet

Student's Name: TEH JUN HAO **Date:** 22 July 2026 12 pm **Meeting No:** 6

Project tittle:

CENTRALIZED SECURITY CONFIGURATION SCANNING TOOL TO DETECT RANSOMWARE-RELATED MISCONFIGURATIONS IN SME ENVIRONMENTS

Intake:

Supervisor's Name: Dr. Hassan Jamil Syed **Supervisor's signature**

Item for discussion (Noted by student before mandatory supervisory meeting):

1.Final Year Project (FYP) progress overview

2. Demonstrate System

Record of discussion (noted by student during mandatory supervisory meeting):

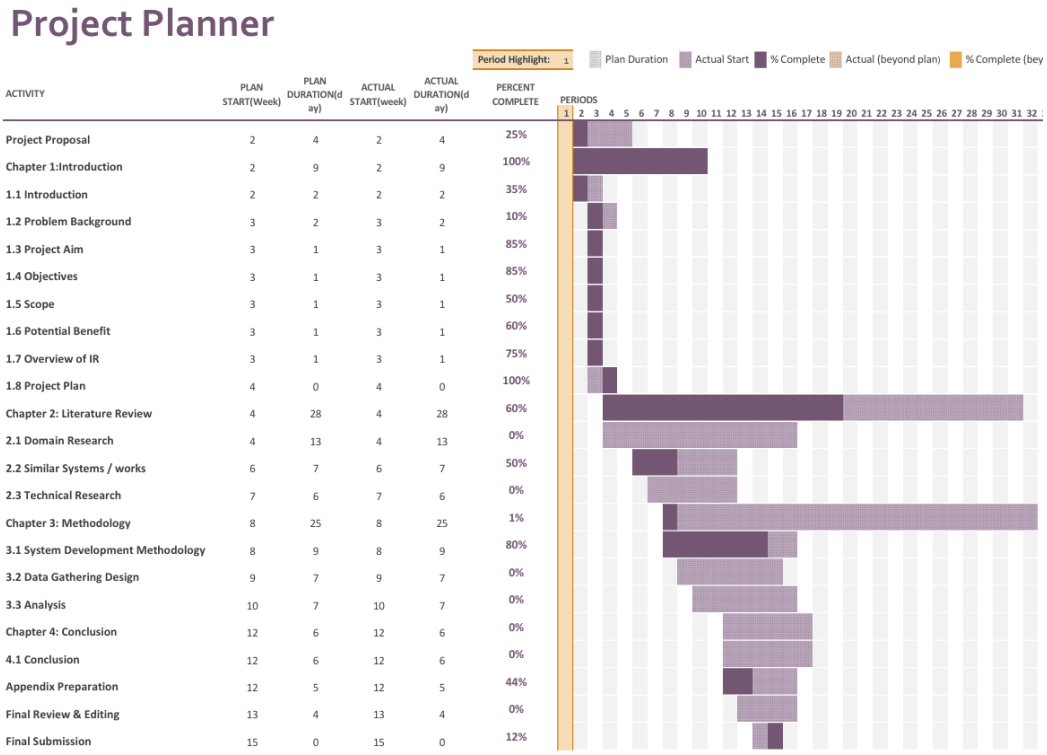
1.Student presented overall FYP progress.

2.Supervisor reviewed the system and provided feedback on clarity and completeness.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

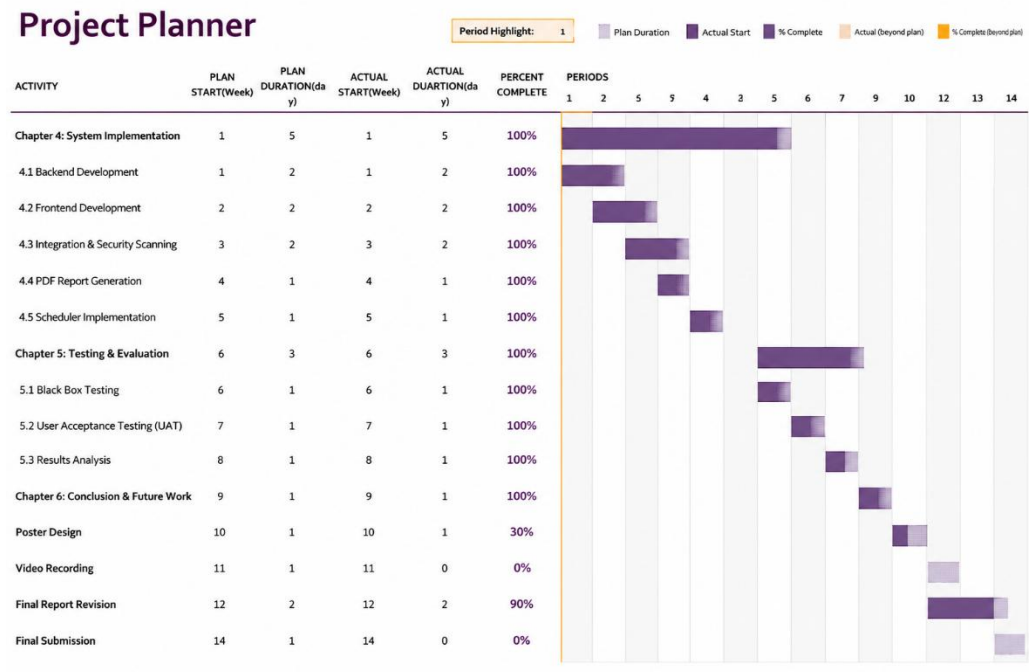
Figure 43: Meeting Log Sheets page 6

Appendices E: Gantt Chart



ACTIVITY	PLAN START(Week)	PLAN DURATION(day)	ACTUAL START(week)	ACTUAL DURATION(day)	PERCENT COMPLETE
Project Proposal	2	4	2	4	25%
Chapter 1:Introduction	2	9	2	9	100%
1.1 Introduction	2	2	2	2	35%
1.2 Problem Background	3	2	3	2	10%
1.3 Project Aim	3	1	3	1	85%
1.4 Objectives	3	1	3	1	85%
1.5 Scope	3	1	3	1	50%
1.6 Potential Benefit	3	1	3	1	60%
1.7 Overview of IR	3	1	3	1	75%
1.8 Project Plan	4	0	4	0	100%
Chapter 2: Literature Review	4	28	4	28	60%
2.1 Domain Research	4	13	4	13	0%
2.2 Similar Systems / works	6	7	6	7	50%
2.3 Technical Research	7	6	7	6	0%
Chapter 3: Methodology	8	25	8	25	1%
3.1 System Development Methodology	8	9	8	9	80%
3.2 Data Gathering Design	9	7	9	7	0%
3.3 Analysis	10	7	10	7	0%
Chapter 4: Conclusion	12	6	12	6	0%
4.1 Conclusion	12	6	12	6	0%
Appendix Preparation	12	5	12	5	44%
Final Review & Editing	13	4	13	4	0%
Final Submission	15	0	15	0	12%

Figure 44: Gantt Chart Sem1



ACTIVITY	PLAN START(Week)	PLAN DURATION(day)	ACTUAL START(Week)	ACTUAL DUARTION(day)	PERCENT COMPLETE
Chapter 4: System Implementation	1	5	1	5	100%
4.1 Backend Development	1	2	1	2	100%
4.2 Frontend Development	2	2	2	2	100%
4.3 Integration & Security Scanning	3	2	3	2	100%
4.4 PDF Report Generation	4	1	4	1	100%
4.5 Scheduler Implementation	5	1	5	1	100%
Chapter 5: Testing & Evaluation	6	3	6	3	100%
5.1 Black Box Testing	6	1	6	1	100%
5.2 User Acceptance Testing (UAT)	7	1	7	1	100%
5.3 Results Analysis	8	1	8	1	100%
Chapter 6: Conclusion & Future Work	9	1	9	1	100%
Poster Design	10	1	10	1	30%
Video Recording	11	1	11	0	0%
Final Report Revision	12	2	12	2	90%
Final Submission	14	1	14	0	0%

Figure 45: Gantt Chart sem2

Appendices F: Respondent Demographic Profile

Participant	**Gender**	**Age**	**Occupation**
Alex	Male	18–24	IT Student
Brian	Male	18–24	IT Student
Daniel	Male	18–24	IT Student
Kevin	Male	25–34	System Engineer
Jason	Female	25–34	IT Support
Ryan	Female	25–34	IT Support
Ethan	Male	45 and above	System Engineer
Samuel	Male	45 and above	IT Student
Aaron	Male	45 and above	IT Student
Lucas	Male	45 and above	Cybersecurity Personnel
Chloe	Female	25–34	Cybersecurity Personnel
Sophia	Male	18–24	System Engineer
Olivia	Male	35–44	Cybersecurity Personnel
Emma	Female	25–34	IT Student
Mia	Female	18–24	IT Administrator
Ava	Female	45 and above	IT Administrator
Isabella	Male	35–44	IT Student
Grace	Male	35–44	Cybersecurity Personnel
Lily	Male	18–24	Cybersecurity Personnel
Hannah	Male	25–34	IT Support

Zoe	Male	25–34	Cybersecurity Personnel
Ella	Male	35–44	Cybersecurity Personnel
Noah	Male	35–44	System Engineer
Liam	Female	25–34	IT Support
Mason	Male	18–24	IT Support
Logan	Male	35–44	Cybersecurity Personnel
Jacob	Male	35–44	IT Administrator
Benjamin	Female	25–34	Cybersecurity Personnel
William	Female	45 and above	IT Student
James	Male	18–24	System Engineer
Michael	Male	35–44	Cybersecurity Personnel
Alexander	Male	25–34	System Engineer
Daniel K	Male	35–44	System Engineer
Matthew	Male	45 and above	IT Student
Andrew	Male	45 and above	Cybersecurity Personnel
Sophia L	Female	35–44	System Engineer

Figure 46: Respondent Demographic Profile

Appendices G: Source Code (Important Only)

```
@app.route("/")
def dashboard():
    records = ensure_summary_fields(load_json(RESULTS_FILE, []))
    saved_targets = load_json(TARGETS_FILE, [])
```

```
@app.route("/scan", methods=["POST"])
def scan():
    target = {
        "os_type": request.form.get("os_type", "linux"),
        "ip": request.form.get("ip", "").strip(),
        "username": request.form.get("username", "").strip(),
```

```
@app.route("/result/<record_id>")
def view_result(record_id):
    records = ensure_summary_fields(load_json(RESULTS_FILE, []))
    record = next((r for r in records if r["id"] == record_id), None)
    if not record:
        flash("Result not found.", "error")
```

Figure 47:Flask routing implementation

```
def connect_ssh(host: str, username: str, password: str):
    client = paramiko.SSHClient()
    client.set_missing_host_key_policy(paramiko.AutoAddPolicy())
    client.connect(hostname=host, username=username, password=password, timeout=10)
    return client
```

Figure 48: Linux Remote Scanning Module

```
def scan_windows(ip, username, password) -> List[Dict]:
    if winrm is None:
        raise ImportError("pywinrm is not installed. Install it using: pip install pywinrm")

    print(f"\n[*] Connecting to Windows target {ip} using WinRM...")
    session = winrm.Session(
        f"http://{ip}:5985/wsman",
        auth=(username, password)
    )
    print("[+] Connected successfully.")
```

Figure 49: Windows remote scanning using WinRM

D: > ME > APU > FYP > fyp_web_dashboard > app.py > calculate_risk_score

```python
def calculate_risk_score(results):
    Produces a normalized 0-100 risk score.
    0 means no detected configuration risk, 100 means all checks are high-risk failures.
    """
    if not results:
        return 0, "Unknown"

    max_possible = len(results) * 20
    raw_score = sum(result_risk_points(r) for r in results)
    score = round((raw_score / max_possible) * 100)
    score = max(0, min(score, 100))

    if score >= 70:
        level = "High"
    elif score >= 40:
        level = "Medium"
    else:
        level = "Low"
    return score, level

```

Figure 50: Risk score calculation algorithm

```python
def generate_pdf_report(record):
    buffer = BytesIO()
    doc = SimpleDocTemplate(
        buffer,
        pagesize=landscape(A4),
        rightMargin=12 * mm,
        leftMargin=12 * mm,
        topMargin=12 * mm,
        bottomMargin=12 * mm,
    )

    styles = getSampleStyleSheet()
    title_style = ParagraphStyle(
        "TitleCenter"
```

Figure 51: PDF report generation module

```python
def scheduler_loop():
    while True:
        if scheduler_state["enabled"]:
            scheduler_state["status"] = "Running"
            saved_targets = load_json(TARGETS_FILE, [])
            for target in saved_targets:
                try:
                    run_scan_for_target(target, source="Scheduled")
                except Exception as e:
                    error_result = [{
                        "target_ip": target.get("ip", "Unknown"),
                        "os": target.get("os_type", "Unknown"),
                        "check": "Scheduled Scan Error",
                        "status": "INFO",
                        "risk": "MEDIUM",
                        "details": str(e),
                        "remediation": "Verify target IP, credentials, network connectivity, SSH/WinRM service, and firewall rules."
                    }]
                    save_scan_record(target, error_result, source="Scheduled")

            scheduler_state["last_run"] = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
            interval_seconds = max(1, int(scheduler_state["interval_minutes"])) * 60
            scheduler_state["next_run"] = datetime.fromtimestamp(time.time() + interval_seconds).strftime("%Y-%m-%d %H:%M:%S")
            time.sleep(interval_seconds)
        else:
            scheduler_state["status"] = "Stopped"
            time.sleep(2)
```

Figure 52: Scheduled scanning implementation